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Inhomogeneous dust collapse in Painlevé-Gullstrand coordinates

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ENDORSEMENT

This is to certify that this undergraduate thesis entitled **Inhomogeneous dust collapse in Painlevé-Gullstrand coordinates**, prepared and submitted by Ryan Anthony G. Maghuyop, is hereby endorsed for acceptance in partial fulfillment of the requirements for the degree of Bachelor of Science in Physics.

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ABSTRACT

INHOMOGENEOUS DUST COLLAPSE IN PAINLEVÉ-GULLSTRAND COORDINATES

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We investigate the gravitational collapse of inhomogeneous dust distributions using Painlevé–Gullstrand (PG) coordinates. This coordinate choice enables seamless matching between the local collapsing matter and an external universe. Moreover, PG coordinates are horizon-penetrating, allowing us to probe the geometry of collapse beyond the formation of apparent horizons. Using this formalism, we consider three matter configurations: homogeneous dust in vacuum, inhomogeneous dust in vacuum, and inhomogeneous dust in a collapsing universe.

A key insight is that local matter inhomogeneity delays collapse and influences apparent horizon behavior, which is a result we conjecture to hold regardless of the specific inhomogeneity profile. By parametrizing the initial matter density via its central density ρ_c and boundary r_b , we show that collapse into singularities is qualitatively insensitive to these parameters, though apparent horizon formation is not. For the vacuum exterior case, we characterize the full parameter space and derive analytic expressions for horizon formation times. We also find that embedding the local matter in a collapsing background universe accelerates the collapse. Due to numerical limitations, a complete analysis of this case remains an open direction for future work.

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Chapter 1

Introduction

1.1 Background of Study

Gravitational collapse of matter is a cornerstone phenomenon in general relativity, governing the formation of stable astrophysical objects such as stars and white dwarfs, as well as exotic end-states like neutron stars and black holes. Traditionally, collapse models or structure formation studies roughly fall into one of three methodological categories: (1) exact analytic models such as the Oppenheimer–Snyder solution [1], which describes a spherically symmetric, homogeneous dust cloud collapsing in a vacuum; (2) perturbative approaches, where small density fluctuations evolve within an expanding Friedmann–Lemaître–Robertson–Walker (FLRW) universe, typically studied through linearized Einstein equations and scalar perturbations [2–6]; and (3) large-scale N-body simulations (e.g. Millenium simulation), which track the evolution of matter under gravity using Newtonian or post-Newtonian approximations on cosmological scales [7–9].

While each of these approaches offers valuable insights, they also come with limitations. The Oppenheimer–Snyder model, for instance, assumes perfect homogeneity and ignores the influence of a cosmological background. Perturbative methods restrict the analysis to small deviations and fail to capture the fully nonlinear regime of collapse. N-body simulations, on the other hand, often rely on Newtonian gravity and require significant computational resources, while making assumptions that obscure the detailed spacetime geometry of collapse.

A full, exact, and nonlinear treatment of the collapse of spherically symmetric, inhomogeneously distributed matter—particularly one that accounts for its embedding within a cosmological spacetime—remains analytically challenging and relatively

unexplored. Such a framework is essential for bridging the gap between idealized collapse and realistic structure formation, and for understanding how local gravitational dynamics are shaped or modified by the exterior universe.

This motivates the search for exact or semi-analytic inhomogeneous solutions to Einstein’s field equations (EFEs) that can describe the decoupling of local overdensities from the global expansion, and their subsequent collapse into compact objects. To study such scenarios numerically, it is essential to formulate Einstein’s equations in a coordinate system that is both regular across horizons and suitable for numerical evolution. This ensures that simulations can proceed beyond the formation of apparent horizons and into the strong-field regime without encountering coordinate singularities. The Lemaitre-Tolman-Bondi (LTB) models in Painlevé-Gullstrand (PG) coordinates provide a promising framework for modeling such systems, offering a physically motivated foliation of spacetime with coordinates that remain regular at horizons.

1.2 Statement of the Problem

A central challenge in gravitational physics is to understand how local overdensities evolve and collapse in the presence of a surrounding universe. While classical models such as the Oppenheimer-Snyder solution offer insight into collapse in vacuum, and perturbative and numerical methods capture aspects of structure formation in expanding cosmologies, there remains a critical gap in fully nonlinear and relativistically consistent treatments of collapse embedded within a cosmological context.

Specifically, it is unclear how the presence of an external universe—particularly one that is itself evolving—modifies the dynamics of local matter collapse. Existing exact solutions typically assume idealized conditions such as homogeneity, pressureless matter, or vacuum exteriors, while large-scale simulations often trade relativistic accuracy for computational feasibility. Moreover, most analytic models fail to provide a unified framework in which local gravitational collapse and global cosmic evolution can be treated self-consistently and without coordinate singularities.

This study attempts to address this gap by exploring the collapse of inhomogeneous dust within the Painlevé-Gullstrand formulation of the LTB spacetime, which maintains regularity across horizons and allows for numerical evolution beyond singularity formation. While the chosen formalism supports the embedding of local

collapse into a larger cosmological background, it imposes important constraints: only vacuum and collapsing external universes can be treated exactly, and dynamical interplay with expanding backgrounds remains out of reach. Despite these limitations, this work aims to provide qualitative and quantitative insight into how initial inhomogeneities and the surrounding universe influence the onset of singularity and apparent horizon formation with the collapsing matter.

1.3 Scope and Limitation of the Study

This work is confined to spherically symmetric dust spacetimes and does not incorporate pressure, rotation, or more complex forms of matter. The collapse is modeled within a $3+1$ LTB spacetime expressed in PG coordinates, which are chosen for their numerical regularity and ability to penetrate horizons. The models studied include both homogeneous and radially inhomogeneous energy density profiles, allowing us to examine the causal structure and horizon dynamics resulting from different initial matter configurations.

While the PG-LTB formalism enables the embedding of collapsing regions within a larger cosmological background, this is limited in two important ways. First, the global spacetime into which the collapse is embedded must either be a vacuum or itself be collapsing; truly expanding backgrounds (as in realistic cosmological models) cannot be incorporated without significant modification of the formalism. Second, although the study includes matched LTB-FLRW spacetimes, these rely on a simplified and idealized transition rather than a fully dynamical interplay between local collapse and cosmic expansion. Therefore, while the framework can model local collapse in a vacuum or in a collapsing FLRW universe, it does not capture the full range of possibilities involved in structure formation in an expanding universe. Nonetheless, the results provide a foundational step toward future work that may extend these models to more general, fully dynamical scenarios.

1.4 Outline of the manuscript

This manuscript contains seven (7) chapters. Chapter 2 provides the reader with the necessary mathematical preliminaries in general relativity to understand gravitational collapse studies. In addition to textbook topics such as structures on manifolds, geodesics, and the Einstein field equations, we also discuss the ADM formalism as a tool for dynamically evolving systems in a covariant theory like general relativity, which does not admit a global definition of time. Moreover, we introduce apparent horizons as tools in numerical relativity for probing the existence of black holes and singularities in spacetime.

In Chapter 3, we review standard solutions to the Einstein field equations such as the Schwarzschild and FLRW metrics, and present an alternative coordinate system for these solutions known as Painlevé–Gullstrand coordinates. We also introduce the lesser-known Lemaitre–Tolman–Bondi (LTB) solution, which describes spherically symmetric, inhomogeneous matter sources relevant to our study. We then present and rederive a solution originally found by Lasky and Lun [10], which we use for our simulations.

We discuss the results and analysis of our simulations in Chapters 5 and 6, where we embed our collapsing matter sources in a vacuum universe and in a collapsing FLRW background, respectively. Finally, Chapter 7 provides a summary of our results and offers some recommendations for future work.

Chapter 2

Basic elements of general relativity

For this chapter, we shall establish some of the key elements of Einstein’s theory of general relativity that are necessary for this thesis. Some references that might be useful as supplementary readings are [11–15]. Throughout this text, we adopt a +2 Lorentzian signature for the metric tensor. We shall also use the abstract index notation, which differentiates the tensor g_{ab} (labeled by Latin letters) from its components $g_{\alpha\beta}$ (labeled by Greek letters). These indices run from 0 to 3, while the middle Latin letters $i, j, k, \dots$ will run from 1 to 3, usually reserved for spatial indices. We adopt the Einstein summation convention of summing repeated indices. Lastly, we shall use geometrized units and set the gravitational constant and speed of light to unity, i.e. $G = c = 1$. This choice leaves all physical variables to be in units of length, and we shall use the solar mass $M_\odot \approx 1.5 \text{ km}$ as the characteristic length scale for our simulations later on.

2.1 Geometry of spacetime

The fundamental object of concern in general relativity is the *spacetime* that we live on. Mathematically, a spacetime is described by a pair $(\mathcal{M}, g_{ab})$, where $\mathcal{M}$ is a four-dimensional, smooth, C^∞ manifold, and g_{ab} is a Lorentzian metric tensor defined on $\mathcal{M}$. The elements of $\mathcal{M}$ are called events. The topology of $\mathcal{M}$ provides a notion of events being “close” to one another, allowing for continuity and differentiability. Moreover, $\mathcal{M}$ is covered by a collection of open sets $\{\mathcal{U}_\alpha\}$ such that $\mathcal{M} = \bigcup_\alpha \mathcal{U}_\alpha$. Each open set is equipped with a *chart* $(\mathcal{U}_\alpha, \varphi)$, where $\varphi : \mathcal{U}_\alpha \rightarrow x^\alpha \subseteq \mathbb{R}^4$ is a smooth mapping that assigns four real parameters to the events in $\mathcal{U}_\alpha$. On overlapping regions, the transition maps between coordinate charts are required to be C^∞ differentiable.

In essence, what all of this mathematical formalism really is trying to say is that every point in the manifold can be locally coordinatized or labeled by four parameters depending on where it is on the manifold (i.e. which open set the event belongs to). Hence, the map φ is what we commonly call the *coordinate system* with its components given by the coordinate functions x^a . At a point $p \in \mathcal{M}$, these coordinate functions locally define a linearly independent set of vectors $\{\partial_a\} \equiv \{\frac{\partial}{\partial x^a}\}$ known as the coordinate basis, which spans the tangent space $\mathcal{T}_p\mathcal{M}$ at p .

It is crucial to emphasize that coordinates are arbitrary labels. The physical laws that we observe are not tied to any particular coordinate system. Rather, they are expressed using tensors which are geometric objects that are invariant to the choice of coordinates. The true physical content lies in the tensorial structure of the manifold, particularly in the metric tensor g_{ab} .

The metric tensor g_{ab} is an $(0, 2)$ -tensor that assigns the definition of length in the spacetime manifold. We can think of the metric tensor as an inner product operator between two vectors X^a and Y^a living on the manifold, where

$$\vec{X} \cdot \vec{Y} \equiv g_{ab}X^aY^b. \quad (2.1)$$

From this inner product, we can define the magnitude of a vector $|\vec{X}| = \sqrt{g_{ab}X^aX^b}$ and angles between two vectors $\cos \theta = |\vec{X} \cdot \vec{Y}|/(|\vec{X}||\vec{Y}|)$ ¹. This allows us to define *orthogonality* between vectors by the condition $g_{ab}X^aY^b = 0$. Consequently, we are led to the concept of an *orthonormal basis* set $\{e_{(\mu)}^a\}$, in which the basis vectors satisfy the relation

$$g_{ab}e_{(\mu)}^ae_{(\nu)}^b \equiv g_{\mu\nu} = \eta_{\mu\nu}, \quad \text{or} \quad \mathbf{g}(\vec{e}_{(\mu)}, \vec{e}_{(\nu)}) = \eta_{\mu\nu} \quad (2.2)$$

where $g_{\mu\nu}$ are the components of the metric in some basis² and $\eta_{\mu\nu}$ is the typical invariant Minkowski metric of flat spacetime. It is important to note that this equality only holds at a point in a locally inertial frame. In Cartesian coordinates, $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$. Essentially, we can think of this operation as supplying an $(0, 2)$ -tensor with two $(1, 0)$ -tensors (vectors) to produce a $(0, 0)$ -tensor (scalar). Now that we know how the metric acts on basis vectors, we also understand how it acts on any arbitrary vector. Revisiting its inner product definition,

¹For this section only, boldface characters denote entire vectors, one-forms, or tensors (as geometric objects), for illustrative purposes.

²For all intents and purposes, we can safely treat $g_{\mu\nu}$ as simply a 4×4 matrix.

$$\begin{aligned}\vec{\mathbf{X}} \cdot \vec{\mathbf{Y}} &= g_{ab} (X^\mu e_{(\mu)}^a) (Y^\nu e_{(\nu)}^b) \\ &= g_{\mu\nu} X^\mu Y^\nu,\end{aligned}\tag{2.3}$$

which is manifestly a scalar. Usually, the orthonormal basis set chosen is the coordinate basis $\{\partial_\mu\}$, which are unit vectors pointing in the direction of the coordinate x^μ . In general, a set of orthonormal basis vectors constitutes a coordinate basis if the basis vectors commute, which holds here since partial derivatives commute.

We can also instead supply the metric with one vector to define a one-form ω_a :

$$\omega_a \equiv g_{ab} \omega^b \tag{2.4}$$

Arbitrary one-forms can also be written as a linear combination of components and orthonormal basis one-forms $\{e^\mu\}$ such that $\omega_a = \omega_\mu e_a^{(\mu)}$. We can think of this definition of one-forms as an $(0, 2)$ -tensor being supplied a $(1, 0)$ -tensor to produce a $(0, 1)$ -tensor. The metric thus provides a natural isomorphism between the vector space $\mathcal{V}$ (with $\omega^a \in \mathcal{V}$) and the covector space $\mathcal{V}^*$ (with $\omega_a \in \mathcal{V}^*$). Accordingly, there exists an inverse metric g^{ab} which relates one-forms back to vectors such that $\omega^a = g^{ab} \omega_b$. We call this property of the metric/inverse metric as *index lowering/raising*. Similarly, the inverse metric acts on the basis one-forms as

$$g^{ab} e_a^{(\mu)} e_b^{(\nu)} \equiv g^{\mu\nu} = \eta^{\mu\nu}, \quad \text{or} \quad \mathbf{g}(\mathbf{e}^\mu, \mathbf{e}^\nu) = \eta^{\mu\nu} \tag{2.5}$$

and we are now equipped to understand how the inverse metric acts on arbitrary one-forms. From the coordinate basis vector set $\{\partial_\mu\}$, we can also define a dual basis of one-forms $\{dx^\mu\}$ such that

$$dx^\mu(\partial_\nu) = \partial_\nu(dx^\mu) \equiv \delta_\nu^\mu. \tag{2.6}$$

where δ_ν^μ is the Kronecker delta. Geometrically, the basis dx^μ are level sets of the coordinate x^μ or surfaces of constant x^μ . In this coordinate basis, the metric tensor $g_{\mu\nu}$ defines the *line element* ds^2 , or the squared invariant interval between infinitesimally nearby events, as

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \tag{2.7}$$

³Similarly, we can simply treat its components $g^{\mu\nu}$ as the inverse matrix of $g_{\mu\nu}$.

2.2 Derivatives in a manifold

Up to this point, we have described how spacetime can be equipped with tensors and how the metric tensor provides a means to measure lengths, angles, and distances. However, a complete understanding of spacetime geometry requires us to understand how these objects change from point to point. In particular, if we wish to compare tensor fields at different points on the manifold, we must introduce tools that allow us to differentiate tensors in a way that respects the underlying geometric structure of spacetime.

Lie Derivative

The most natural way to describe the change of a tensor field along the flow of a vector field is through the *Lie derivative*. Suppose we have a smooth vector field X^a on the manifold. This vector field generates a flow that “pushes” points along the integral curves of X^a . Given a tensor field $\mathbf{T}$, the Lie derivative $\mathcal{L}_X \mathbf{T}$ tells us how $\mathbf{T}$ changes as we move along this flow.

At an intuitive level, the Lie derivative answers the question: *How does a tensor field change as I drag it along the direction of a vector field?* For the scalar field ψ , its Lie derivative along X^a is simply the directional derivative:

$$\mathcal{L}_X \psi = X^\mu \partial_\mu \psi. \quad (2.8)$$

If $\mathcal{L}_X \psi = 0$, then we say that ψ is a conserved quantity for observers that move along the integral curves of X^a ⁴. While the Lie derivative provides a natural way to track how fields change along flows, it lacks the ability to compare vectors or tensors at different points in a geometrically meaningful way. In curved spacetime, where the notion of “sameness” across points is nontrivial due to curvature, we require a more refined tool: one that can account for how tensor components change and how the underlying basis vectors themselves shift across the manifold. This leads us to the covariant derivative.

⁴For the purposes of this thesis, we only care about Lie derivatives of scalars. For general tensor fields, the Lie derivative acts as a generalized commutator.

Covariant Derivative

Unlike the Lie derivative, which is intrinsic to the manifold's differentiable structure, the *covariant derivative* requires the additional specification of a *connection* C_{bc}^a . Unlike $\mathbb{R}^n$ where there is an intuitive notion of “sliding” to compare vectors with different origin points, general manifolds do not have a trivial way of *parallel transporting* vectors at infinitesimally separated points. The choice of connection endows us with a rule for parallel transport. On a manifold equipped with a metric, the natural choice is the *Levi-Civita connection*, which satisfies two key properties:

1. *Metric compatibility*: $\nabla_c g_{ab} = 0$, meaning the inner product between vectors is preserved under parallel transport.
2. *Torsion-free*: The connection is symmetric in its lower indices: $C_{bc}^a = C_{cb}^a$.

The covariant derivative $\nabla_a T^{b\dots}_{c\dots}$ of a tensor field $T^{b\dots}_{c\dots}$ generalizes the idea of a directional derivative to curved spacetime. It respects the tensorial nature of the object being differentiated and accounts for changes not just in the components of the tensor, but also in the basis vectors themselves. Explicitly,

$$\nabla_c T_b^a = \partial_c T_b^a + \Gamma_{dc}^a T_b^d - \Gamma_{bc}^d T_d^a, \quad (2.9)$$

where $\Gamma_{bc}^a = \frac{1}{2}g^{ad}(\partial_c g_{db} + \partial_b g_{dc} - \partial_d g_{bc})$ are the Christoffel symbols. When the partial derivatives of the metric components g_{ab} vanish (as in flat spacetime), the Christoffel symbols Γ_{bc}^a are zero, and the covariant derivative reduces to the ordinary partial derivative. As a direct consequence of these extra terms, the covariant derivative of an arbitrary tensor T_d^c does not commute in a general curved manifold:

$$[\nabla_a, \nabla_b]T_d^c \neq 0. \quad (2.10)$$

This non-commutativity is a manifestation of spacetime curvature. Intuitively, curvature measures the failure of tensors to return to their original form when parallel transported around a closed loop. This failure arises from the fact that the basis vectors themselves vary from point to point in a curved geometry. From this fact, we obtain the *Riemann tensor* R_{bcd}^a defined as

$$[\nabla_a, \nabla_b]V_c = V_d R_{cab}^d, \quad (2.11)$$

or alternatively,

$$R_{bcd}^a = \partial_c \Gamma_{bd}^a - \partial_d \Gamma_{bc}^a + \Gamma_{ec}^a \Gamma_{bd}^e - \Gamma_{ed}^a \Gamma_{bc}^e. \quad (2.12)$$

The Riemann tensor has a number of symmetries and identities, most importantly being the Bianchi identities:

$$\nabla_e R_{abcd} + \nabla_d R_{abec} + \nabla_c R_{abde} = 0. \quad (2.13)$$

It has a unique trace $R_{ab} = R_{acb}^c$ called the (symmetric) Ricci tensor, which itself has a unique scalar trace $R = R_a^a$ called the Ricci scalar. Using these two objects, we can define the symmetric Einstein tensor G_{ab} :

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R. \quad (2.14)$$

It can be shown that its covariant derivative $\nabla^a G_{ab} = 0$ vanishes as a consequence of the Bianchi identities.

2.3 Geodesics

One of the key paradigm shifts that the theory of relativity brought upon is the idea that the acceleration due to gravitational “force” is, in fact, a manifestation of the underlying geometry of curved spacetime. Einstein posited that inertial observers are ones that are *freely falling* or moving in the “straightest” possible paths called *geodesics*. For a freely-falling observer following a worldline $x^a(\lambda)$ where λ is an affine parameter along the geodesic curve γ , its extremal paths are given by the geodesic equation:

$$\frac{d^2 x^a}{d\lambda^2} + \Gamma_{bc}^a \frac{dx^b}{d\lambda} \frac{dx^c}{d\lambda} = 0. \quad (2.15)$$

This equation describes how the observer’s path is governed entirely by the space-time geometry. It is equivalent to the condition that the tangent vector $n^a = dx^a/d\lambda$ is parallel transported along itself:

$$n^b \nabla_b n^a = 0. \quad (2.16)$$

In other words, a freely falling particle follows a path such that its four-velocity remains unchanged under parallel transport, reflecting the absence of any “true” force

acting on it. These two expressions also represent the four-acceleration of the observer. That is, the left-hand side of either equation corresponds to the observer's deviation from inertial motion. Therefore, if either expression is zero, it means the observer experiences no proper acceleration, i.e. no force is acting on them. In the context of general relativity, this is precisely what we interpret as the absence of gravitational forces: free-falling observers move along geodesics and feel no acceleration.

This leads us naturally to the question: if freely falling observers experience no proper acceleration, how do we account for the relative acceleration between two such nearby observers? In flat spacetime, two initially parallel geodesics remain parallel. However, in curved spacetime, tidal effects can cause nearby geodesics to converge or diverge. This relative acceleration is encoded in the Riemann tensor via the geodesic deviation equation.

Consider a congruence of timelike geodesics parametrized by λ . Let s be a continuous parameter that distinguishes between neighboring geodesics such that our family of geodesics can be labeled as $x^a(\lambda; s)$. The deviation vector $\xi^a \equiv \partial x^a / \partial s$ then describes the difference between two infinitesimally separated geodesics. The relative acceleration of these nearby freely falling trajectories is governed by the geodesic deviation equation:

$$\xi^c \nabla_c (\xi^b \nabla_b n^a) = -R^a_{bcd} n^b \xi^c n^d \quad (2.17)$$

The right-hand side defines the *tidal tensor*, which describes how the curvature of spacetime stretches or squeezes the local neighborhood of the observer. This is how gravity manifests in general relativity: not as a force acting on a single particle, but as differential accelerations between neighboring geodesics due to spacetime curvature⁵.

So far, our discussion has focused on timelike geodesics. That is, the tangent vectors are normalized to $n^a n_a = -1$ and describes only the freely falling trajectories of massive particles. Another key aspect of general relativity is that light itself is affected by gravity despite its massless nature. As such, massless particles like photons also follow geodesics, called *null geodesics*. The tangent vector to these geodesics k^a satisfies the relation $k^a k_a = 0$. Despite this key difference, similar relations can be derived and the null geodesic equation is given by

⁵A more pedagogical derivation and explanation can be found in Chapter 1.10 of [14].

$$k^b \nabla_b k^a = 0. \quad (2.18)$$

Thus, the paths of light rays and gravitational waves are governed by the geometry of spacetime in the same way as those of massive particles though they probe a different causal structure via null curves.

The causal structure of a spacetime is determined by the behavior of null geodesics, which generate the local light cones at each point. These light cones encode the causal relations between events, determining which events can influence or be influenced by a given event. A spacetime is said to contain a *singularity* if it is geodesically incomplete; i.e., there exists inextendible geodesics whose affine parameter takes only a finite range. This provides an operational and coordinate-independent definition of singularities.

As seen in (2.17), the deviation vector ξ^a between nearby geodesics evolves under the influence of curvature. In highly curved spacetimes, neighboring geodesics may converge and intersect at caustics, where $\xi^a \rightarrow 0$. These intersections mark the breakdown of the geodesic congruence and the deviation vector becomes ill-defined. Thus, one way to characterize singularities is by identifying the regime beyond which ξ^a is no longer valid, often due to the formation of caustics.

2.4 Einstein field equations and matter content

We now know that what we measure as the gravitational field is simply the curvature or geometry of the underlying spacetime. This curvature is sourced or generated by the collection of energy, momentum, and stress present in the system. All contributions of mass-energy in the spacetime is collected in the stress-energy tensor T_{ab} . The relation between the curvature and matter content of the spacetime is expressed in the Einstein field equations (with physical units), given by

$$G_{ab} = \frac{8\pi G}{c^4} T_{ab}. \quad (2.19)$$

This relation states that there is an inherent coupling between the latent geometry and the existing matter. Moreover, keeping the magnitudes of the physical constants in mind, this relation states that an extremely large amount of matter will generate a tiny amount of curvature. By virtue of the Bianchi identities (2.13), the covariant

derivative of the stress-energy tensor must also necessarily vanish

$$\nabla_a T^{ab} = 0 \quad (2.20)$$

This equation is known as the *conservation of the stress-energy tensor*, which is a statement that the total amount of mass-energy in a spacetime is conserved. Usually, these equations are much simpler than the Einstein field equations and are used as supplementary relations to constrain the metric functions.

As a first approximation, almost all matter sources in general relativity can be assumed to be a *perfect fluid*, which has the form

$$T^{ab} = (\rho + P)n^a n^b + P g^{ab}, \quad (2.21)$$

where ρ is the energy density, P is the isotropic pressure, and n^a is the tangent vector to the fluid's worldline. The energy density is effectively a measure of the overall amount of matter in the spacetime, while the pressure determines the self-interaction in the matter source. The choice of isotropic pressure dictates that the matter has no tension or compression forces, i.e., no preferred direction. It also implies no internal friction/viscosity, no resistance to shear, and no net heat flow in or out of the system (adiabatic). It also assumes that each fluid element is in local thermal equilibrium. These assumptions drastically simplify the Einstein equations and are valid as long as the physical system is well-described by a single-component, non-dissipative medium.

In many cases, one finds after writing the field equations explicitly that the system of differential equations for the metric functions $g_{\mu\nu}$, along with the matter variables ρ and P , is underdetermined. To close the system of equations, it is usually sufficient to introduce an extra thermodynamic condition known as an *equation of state (EoS)*, $P = P(\rho)$, that relates the matter variables. In the most general case, thermodynamic equations of state relating the pressure and density are also dependent on the specific entropy of the system. However, this dependence is usually assumed to be very small or negligible for perfect fluids. There are all kinds of equations of state depending on the specific matter source being studied. The simplest one is the linear, barotropic equation of state: $P = \omega\rho$. This can be derived from the ideal gas law, noting that $\rho \propto n/V$.

A special kind of matter source is called *dust*, which is a perfect fluid with negligible pressure ($P = 0$), such that

$$T^{ab} = \rho n^a n^b \quad (2.22)$$

Setting $P = 0$ reduces the number of unknowns in the system and technically is also a valid equation of state on its own. It can be thought of as a linear EoS with $\omega = 0$. Because the pressure vanishes, the matter does not interact with itself completely, and its evolution is determined entirely by gravitation.

2.5 3+1 decomposition and the ADM formalism

From the form of the Riemann tensor (2.12), the Einstein field equations are a system of at most 10 nonlinear, second-order differential equations of the metric tensor components $g_{\mu\nu}$. In the realm of numerical relativity, the Einstein field equations are solved as Cauchy or initial value problems where initial values of “position” $g_{\mu\nu}|_{t=0}$ and “velocity” $\partial_t g_{\mu\nu}|_{t=0}$ are specified. A natural question to ask then is: *How do we define time in general relativity?* Since time is treated as just another parameter in an arbitrary map chosen to label events in the manifold, how do we endow our formalism with the intuitive notion of time that we experience in our daily lives? The mathematical framework that allows this is the *Arnowitt–Deser–Misner (ADM) formalism* [16]. For all but the simplest of astrophysical systems, the Einstein field equations become highly nonlinear differential equations. Hence, the goal of the ADM formalism (also known as the 3 + 1 formalism) is to solve the EFEs for physically realistic dynamical systems using numerical techniques to probe the fundamental aspects of the theory of general relativity in more complicated phenomena.

We start by noting Frobenius’ theorem. This theorem states that if a congruence of timelike curves with tangent vector n^a satisfies the condition

$$n_{[a} \nabla_b n_{c]} = 0, \quad (2.23)$$

then this vector field is everywhere orthogonal to a continuous family of spacelike hypersurfaces. What this means is that the 4D manifold $\mathcal{M}$ can be foliated or split by these 3D spatial surfaces Σ along a timelike flow n^a . Hence, the spacetime admits a clean decomposition into “space + time”, with each hypersurface representing a snap-

shot of space at a given time label t . Note that this ties the definition of coordinate time into being the proper time of the observer whose tangent vector is n^a .

For reasons that will be clear in a while, we define the one-form $\vartheta_a = \nabla_a t$. This one-form defines the level sets of the coordinate function $x^0 = t$ and hence its corresponding vector ϑ^a will be the normal vector pointing in the direction of $+t$. Therefore, the vector ϑ^a is by definition everywhere orthogonal to the family of spatial hypersurfaces $t = \text{constant}$ (hereby named Σ_t), and it can be verified that ϑ_a satisfies Frobenius' theorem. A natural choice for our time direction then would be some vector that is proportional to ϑ^a ; i.e. $n^a \propto \vartheta^a$. Using the metric, we find the norm of this vector:

$$\vartheta_a \vartheta^a \equiv -\frac{1}{\alpha^2}. \quad (2.24)$$

This scalar function α should then modulate the spacing of the $t = \text{constant}$ level sets, and hence can be interpreted as the amount of t that *elapses* between each member of Σ_t . We will return to the discussion of this function later.

The ADM formalism splits the spacetime $(\mathcal{M}, g_{ab})$ into $(\Sigma_t, h_{ab}, K_{ab})$ where h_{ab} is the induced metric defined as

$$h_{ab} = g_{ab} + n_a n_b. \quad (2.25)$$

It is the projection of the 4-metric along each of the slices of Σ_t . Note that despite being a 4-tensor, h_{ab} only has “spatial” components; $h_{00} = h_{0i} = 0$. This tensor captures the intrinsic geometry (i.e. length, angles, etc.) of each spatial slice in the exact same way g_{ab} encodes the geometry of the entire spacetime. The extrinsic curvature⁶ K_{ab} is defined as

$$K_{ab} = -h_a^c h_b^d \nabla_c n_d = -\frac{1}{2} \mathcal{L}_n h_{ab} \quad (2.26)$$

which determines how each spatial slice is embedded into the full spacetime.

⁶Some textbooks refer to this as the second fundamental form.

To highlight the geometric meaning and the need for K_{ab} , let us consider a classic example from differential geometry. Imagine two different embeddings of two-dimensional surfaces: one as a flat Euclidean plane $\mathbb{R}^2$, and the other as a cylinder embedded in three-dimensional space.

Let both surfaces be parametrized by coordinates (x, y) , and suppose that the induced metric on both is the same:

$$h_{ab} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (2.27)$$

This is simply the standard flat metric in two dimensions. Locally, both the plane and the cylinder are indistinguishable from the intrinsic point of view: the induced geometry on the surface is exactly the same. For instance, one can roll a flat sheet of paper into a cylinder without stretching or tearing it. However, the difference between these two surfaces lies in how they are embedded in the higher-dimensional space. The plane is embedded flatly, with its normal vector field constant everywhere. The cylinder, on the other hand, has a normal vector that rotates as one moves along the curved surface. This bending is captured precisely by the *extrinsic curvature* K_{ab} , which measures how the normal vector to the surface changes from point to point.

Under this formalism, the 10 differential equations of the EFEs for the metric components $g_{\mu\nu}$ are transformed into 4 *constraint equations* and 6 *evolution equations*⁷. If we assume spherical symmetry, the 10 ADM equations are effectively reduced to only 4; two constraints (Hamiltonian and momentum constraints) and two evolution equations. To better understand what this means, let us look at the more familiar inhomogeneous Maxwell field equations:

$$\nabla \cdot \vec{\mathbf{E}} = \frac{\rho}{\varepsilon_0} \quad (2.28a)$$

$$\nabla \cdot \vec{\mathbf{B}} = 0 \quad (2.28b)$$

$$\frac{\partial \vec{\mathbf{E}}}{\partial t} = \frac{1}{\mu_0 \varepsilon_0} (\nabla \times \vec{\mathbf{B}} - \mu_0 \vec{\mathbf{J}}) \quad (2.28c)$$

$$\frac{\partial \vec{\mathbf{B}}}{\partial t} = -\nabla \times \vec{\mathbf{E}} \quad (2.28d)$$

The first two equations are time-independent, and hence must be true for any particular instant of time, say $t = t_0$. They *constrain* the possible forms of the vector fields $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ in relation to the matter variables ρ and $\vec{\mathbf{J}}$. We can then impose smooth initial data for the field and matter variables in accordance with our constraints. The

⁷These 10 ADM equations are completely equivalent to the 10 Einstein field equations and there is no loss of information.

remaining two equations then allow us to obtain the values of the two field variables at the next time step $t = t_0 + \Delta t$ using only the initial data that we have imposed. In theory, we can now *evolve* and obtain the entire future of our field variables by integrating forwards in time forever.

In 3 + 1 decomposition schemes, the line element usually admits the form:

$$ds^2 = -\alpha^2 dt^2 + h_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad (2.29)$$

where we introduce the *lapse function* α and the *shift vector* β^i . Notice that on $t = \text{constant}$ hypersurfaces, the spatial 3-metric is simply the spatial components of the induced metric h_{ab} ;

$$ds_\Sigma^2 = h_{ij} dx^i dx^j. \quad (2.30)$$

In a general spherically symmetric spacetime, the lapse function $\alpha(t, r)$ encodes how much proper time elapses along each spatial point on each time slice. If α is found to be strictly a function of time, then it simply becomes a scaling parameter that determines the amount of proper time between two separate time slices. Since the laws of physics are invariant under rescaling in time, we are allowed to find a coordinate transformation:

$$\alpha(t)dt \rightarrow dt' \quad (2.31)$$

that effectively scales α to unity⁸. The shift vector β^i determines how each spatial point will be shifted along each time slice, encoding the difference between the coordinates of an observer moving orthogonally to the hypersurfaces and the coordinates used to label points in space. More concretely, while the lapse function tells us how much proper time elapses between slices, the shift vector tells us how the spatial coordinates of observers “slide” from one time slice to the next. This means that even if two events have the same coordinate label x^i at successive time steps t and $t + \Delta t$, they may not lie along the same normal vector because the coordinates themselves may have shifted sideways across the hypersurface by an amount proportional to $\beta^i \Delta t$.

⁸This “trick” will be done extensively for this thesis.

Therefore, the shift accounts for the fact that the time evolution vector ∂_t is not necessarily aligned with the normal vector n^a to the hypersurfaces. Instead, it decomposes as

$$\partial_t = \alpha n^a + \beta^a, \quad (2.32)$$

where β^a is tangent to the hypersurface. The choice of β^i affects how grid points “move” across the hypersurfaces and is a degree of coordinate freedom that can be exploited to simplify computations or avoid coordinate pathologies.

This entire formalism is the foundation of the field of numerical relativity and one of its most hallmark results is the solution of binary black hole mergers in vacuum. As such, this formalism is too vast to fully elaborate here, and the author refers the interested reader to Chapter 2 of [15] for a comprehensive and pedagogical treatment.

2.6 Trapped surfaces and apparent horizons

When singularities arise in a spacetime as a result of gravitational collapse, it has been conjectured by Roger Penrose [17] that such singularities are always hidden behind an event horizon. However, computing the event horizon of a spacetime requires knowledge of the entire future null infinity which is generally inaccessible, particularly in dynamical spacetimes where the evolution must be integrated step-by-step. In practice, the next best indicator for black hole formation is the detection of an apparent horizon.

To understand the significance of an apparent horizon, consider a congruence of radial null geodesics with tangent vector k^a . We define the expansion scalar

$$\Theta \equiv \nabla_a k^a \quad (2.33)$$

which quantifies the divergence or convergence of the geodesic congruence. When $\Theta > 0$, the null rays diverge, corresponding to outgoing radiation expanding away from the source. Conversely, $\Theta < 0$ indicates that the geodesics are converging, signaling the presence of trapped surfaces. The hypersurface defined by

$$\Theta = 0 \quad (2.34)$$

marks the apparent horizon. At this surface, the null geodesics are momentarily non-expanding—light rays are neither diverging nor escaping, effectively becoming “frozen.”

One limitation of using apparent horizons is that, unlike the event horizon, their existence depends on the choice of foliation; they are *coordinate-dependent*. Whether an apparent horizon is detected can vary with the choice of spacetime slicing. Nevertheless, the event horizon always contains the apparent horizon, and if the spacetime eventually settles into a stationary state, the two horizons coincide exactly [15]. Thus, identifying the apparent horizon provides a practical and accessible method for approximately locating the event horizon in numerical simulations.

Chapter 3

Solutions to the Einstein field equations

We shall now present some special solutions to the Einstein field equations that will be pertinent to understanding the discussions later on. Some supplementary references that the author would recommend are [11, 14, 15, 18, 19].

3.1 Minkowski

The most trivial solution to the EFEs is the Minkowski spacetime, given by the line element

$$ds^2 = -dt^2 + d\vec{x} \cdot d\vec{x}, \quad (3.1)$$

where $\vec{x} = (x, y, z)$ and $\cdot$ is the typical Euclidean dot product. Since all of its metric components are ± 1 , all of its derivatives vanishes and it couples to a vacuum spacetime. This is the natural metric of special relativity and it describes the flat spacetime in the complete absence of gravitation. This line element can also be written in spherical coordinates as

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2, \quad (3.2)$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$ is the standard metric of the unit 2-sphere. It can be verified that despite the dependence of the new metric components on the coordinate functions that the Einstein tensor of this metric still vanishes¹.

¹We expect this since tensorial equalities that hold in one frame holds in all possible frames.

3.2 Schwarzschild

A particularly important and elegant solution to the Einstein field equations is the Schwarzschild solution. It describes the spacetime geometry outside a stationary, spherically symmetric, non-rotating, and uncharged mass distribution. Remarkably, this solution is valid not only for stars and planets, but also for black holes, provided we are interested in the vacuum region outside the source.

The line element of the Schwarzschild solution is given by

$$ds^2 = - \left(1 - \frac{2M_s}{r}\right) dt_s^2 + \left(1 - \frac{2M_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (3.3)$$

where M_s is interpreted as the total (constant) mass of the central object, as seen from spatial infinity. This represents the simplest non-trivial exact solution to the Einstein equations. Computing the Einstein tensor for this metric reveals that this spacetime naturally couples to a vacuum source. Hence, we interpret the fictitious mass generated by the vacuum to be a black hole with an effective mass M_s . Since there is no time dependence, this solution defines an *eternal black hole*; a black hole that has always existed and will always be a black hole. At spatial infinity $r \rightarrow \infty$ or when $M_s = 0$, we note that this solution asymptotically approaches the flat Minkowski metric.

This solution also illustrates the independence of the underlying physics from the choice of coordinates to describe the spacetime. By inspecting the metric, we are led to the conclusion that the spacetime ceases to be regular at $r = 0$ and at $r = 2M_s$. However, it is now a known fact that only $r = 0$ confers a genuine physical singularity while the complication at $r = 2M_s$ is simply a coordinate artifact that can be removed via valid coordinate transformations. It is standard practice to introduce the Eddington-Finkelstein coordinates as well as the Kruskal-Szekeres coordinates as alternative coordinates that cover the Schwarzschild spacetime so that the coordinate singularity at $r = 2M_s$ is made to vanish. Another alternative that has been recognized for its importance recently are the *Painlevé-Gullstrand* coordinates [19, 20].

3.2.1 Painlevé-Gullstrand coordinates of Schwarzschild

The Painlevé-Gullstrand (PG) form of the Schwarzschild metric is

$$ds^2 = -dt^2 + \left(dr + \sqrt{\frac{2M_s}{r}} dt \right)^2 + r^2 d\Omega^2 \quad (3.4)$$

where t is the new PG time coordinate. This metric is manifestly regular at $r = 2M_s$ and the coordinate system smoothly crosses the horizon, in contrast to the stationary Schwarzschild coordinates. One of the distinguishing features of PG coordinates is that the coordinate system is *adapted to the worldlines of radially infalling timelike geodesics*. Specifically, the PG time slices are orthogonal to a congruence of geodesics that start from rest at infinity and fall radially inward under gravity.

An important point to emphasize here is that the $t = \text{constant}$ hypersurfaces are intrinsically flat; i.e. the induced metric h_{ab} on these 3-surfaces is the usual spherical flat metric:

$$ds_\Sigma^2 = dr^2 + r^2 d\Omega^2 \quad (3.5)$$

In ADM terminology, we can recognize from (3.4) that we can interpret this spacetime to have a lapse function $\alpha = 1$ and a radial shift vector component $\beta = \sqrt{2M_s/r}$. We know that the Schwarzschild spacetime is inherently curved, and thus we are led to the conclusion that the nontrivial aspects of the curved spacetime has been pushed into the shift vector. It has been conjectured [19] that all spherically symmetric spacetimes can be recast in Painlevé-Gullstrand type coordinates, barring some unphysical exceptions².

3.3 Friedman-Lemaitre-Robertson-Walker (FLRW) solution

The next solution we consider is a cosmological model, describing the large-scale structure and dynamics of the universe. Since humans have lived for such a short period of time on the cosmological timescale, and our best and most accurate instruments can only probe a limited portion of our past light cone, we have to rely on

²Quoting the cited reference, “... the only obstructions to the global existence of Painlevé-Gullstrand coordinates are the possible existence of wormhole throats, ... and/or negative Misner-Sharp quasi-local mass...”.

philosophical postulates as a starting point in modeling the cosmological spacetime. The point of view adopted by classical cosmology is the *Copernican principle*, which posits that there is no absolute center of the universe, and that every point in space is an equally valid center of everything. Mathematically, this corresponds to the assumption that a valid cosmological spacetime must be homogeneous and isotropic everywhere.

We have also observed that the universe is filled with gravitationally bound baryonic matter such as stars and galaxies, and that there are various phenomena best explained by unobservable components called *dark matter* and *dark energy*. One such phenomenon is the observation that the universe is expanding at an accelerating rate. We have also detected leftover radiation from the cosmic dawn, called the cosmic microwave background, which seems to suggest that the universe started from a point singularity before expanding outward. The signature of this expansion suggests the presence of three types of matter in the universe: baryonic matter, radiation, and non-interacting dark matter. Hence, an appropriate cosmological spacetime model must be able to account for all of these observational facts while at the same time satisfying the EFEs for some kind of cosmological matter source. All of these requirements are satisfied by the Friedmann-Lemaître-Robertson-Walker solution, or FLRW for short.

The FLRW cosmological solution describes a spacetime that has the maximum amount of symmetries in its three spatial directions. It is most commonly presented with the following line element:

$$ds^2 = -d\tau^2 + a(\tau)^2 \left(\frac{d\chi^2}{1 - k\chi^2} + \chi^2 d\Omega^2 \right). \quad (3.6)$$

Here, τ is the proper time measured by the clocks of observers comoving with the expansion of the universe. In other words, an observer that is “riding along” the expansion of the universe will be associated with an $\chi = \text{const.}$ surface throughout the entirety of its history.

Here, $k \in \{-1, 0, 1\}$ indicates the spatial curvature of the spacetime geometry and differentiates the FLRW solution into three disjoint solutions; flat ($k = 0$), closed ($k = 1$), or open ($k = -1$). The sole unknown metric function of this solution is the scale factor $a(\tau)$, which serves to scale the spatial distances as the coordinate time increases. To illustrate this, we look at the $\tau = \text{constant}$ hypersurfaces with the induced metric:

$$ds_{\Sigma}^2 = a(\tau)^2 \left(\frac{d\chi^2}{1 - k\chi^2} + \chi^2 d\Omega^2 \right). \quad (3.7)$$

Hence, an increasing/decreasing scale factor corresponds to an *expanding/collapsing* spatial universe. On the largest cosmological scales, observations indicate that the $k = 0$ FLRW universe provides a sufficiently accurate zeroth-order approximation of the universe [2, 13]. As a coarse-grained model of the universe's matter content, the stress-energy tensor typically coupled to the FLRW solution is that of a linear barotropic perfect fluid³. This represents an idealized cosmic fluid (homogeneous and isotropic) that behaves similarly to an ideal gas permeating all of space. With these choices, the EFEs yield the Friedmann equations⁴:

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi\rho}{3} - \frac{k}{a^2}, \quad (3.8a)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi\rho}{3}(1 + 3\omega) \quad (3.8b)$$

and the conservation equation:

$$\frac{\dot{\rho}}{\rho} = -3(1 + \omega)\frac{\dot{a}}{a} \quad (3.9)$$

For $k = 0$ and $\omega \neq -1$, these admit:

$$\rho(\tau) = \rho(0) \left(1 \pm \tau \sqrt{6\pi(1 + \omega)^2 \rho(0)} \right)^{-2}, \quad (3.10a)$$

$$a(\tau) = A\tau^{\pm \frac{2}{3(1 + \omega)}}, \quad (3.10b)$$

$$\rho \propto a^{-3(1 + \omega)} \quad (3.10c)$$

where A is some constant. Note that because of the squared term on (3.8a), we naturally have two valid solutions corresponding to an expanding and collapsing universe. Based on the observations that the universe is expanding at an accelerating rate, the positive root is usually chosen as the “standard” solution to the Friedmann equations.

³Refer to 2.4 for the exact formulas.

⁴Here, an overdot denotes differentiation with respect to τ .

3.3.1 Painlevé-Gullstrand coordinates of FLRW

Since this is also a spherically symmetric solution, we expect from that this can also be recast in Painlevé-Gullstrand type coordinates. We start by defining the coordinate transformations

$$t = \tau \qquad r = a(\tau)\chi \qquad dr = a d\chi + \chi \dot{a} d\tau \qquad (3.11)$$

which allows us to rewrite the FLRW line element as

$$ds^2 = -d\tau^2 + \frac{1}{1 - kr^2/a^2} \left(dr + \frac{\dot{a}}{a} r d\tau \right)^2 + r^2 d\Omega^2. \qquad (3.12)$$

This time, the $t = \text{constant}$ hypersurfaces have the induced metric:

$$ds_\Sigma^2 = \frac{dr^2}{1 - kr^2/a^2} + r^2 d\Omega^2. \qquad (3.13)$$

In the flat $k = 0$ case, the 3-metric is now stationary and the expansion of the overall spacetime is now encoded on the (positive) radial shift vector $\beta = r\dot{a}/a$.

3.4 Lemaitre-Tolman-Bondi (LTB) dust solution

The Lemaitre-Tolman-Bondi dust solution was originally derived by Lemaitre [21] and further generalized by Tolman [22] and Bondi [23] later on. This family of solutions have been used extensively as an alternative cosmological model to explain the expansion of the universe ([18, 24] and references therein). This model has been explored for its ability to mimic cosmic acceleration without dark energy [25, 26], though subsequent observations (e.g., BAO data) disfavored such explanations.

The LTB solution is a non-stationary and spherically symmetric solution to the Einstein equations coupled to a dust stress energy tensor; $T_{ab} = \rho(\tau, \chi) n_a n_b$. Its line element is given by:

$$ds^2 = -d\tau^2 + \frac{1}{1 + E(\chi)} \left(\frac{\partial R}{\partial \chi} \right)^2 d\chi^2 + R(\tau, \chi)^2 d\Omega^2. \qquad (3.14)$$

Like the FLRW geometry, all observers are comoving. The coordinates are chosen such that all observers in space measure the same proper time, i.e. all clocks are simultaneous, which is evident since $g_{0\alpha} = 0$. Moreover, $g_{00} = -1$ dictates that the time coordinate is also the proper time measured by the comoving observers

between two events that occurred in the spacetime. Since the reference frame is also synchronous, this proper time measurement is also shared by all observers in space [27]. It is easy to show that these observers are freely-falling, since $n^a = (\partial_\tau)^a$ is now a solution to the geodesic equation upon using the coordinate function (and proper time) τ as the affine parameter. Lastly, R is the areal radius, or that the area of the surface defined by $\tau = \text{constant}$ and $\chi = \text{constant}$ centered at the center of symmetry is $4\pi R^2$.

The Einstein equations reduce to two equations:

$$\left(\frac{\partial R}{\partial \tau}\right)^2 = \frac{2M(\chi)}{R} + E(\chi), \quad (3.15a)$$

$$\frac{\partial M}{\partial \chi} = 4\pi\rho R^2 \frac{\partial R}{\partial \chi}. \quad (3.15b)$$

where $M(\chi)$ is another arbitrary function that arises as an integration constant. Equation (3.15a) can be solved for $R(\tau, \chi)$ by integration:

$$\int_0^R \frac{d\tilde{R}}{\sqrt{\frac{2M}{\tilde{R}} + E}} = t - t_{\text{bb}}(\chi). \quad (3.16)$$

We integrate from the event where the areal radius is zero (the big bang) up to some arbitrary R . Since we assume that this solution describes an inhomogeneous universe, we consider the most general case where the coordinate time of the big bang is not the same for different values of the coordinate function χ . This then gives us another arbitrary function, the big bang time $t_{\text{bb}}(\chi)$. From (3.15b), we see that there can exist 2-surfaces where the radial derivative of R vanishes but the radial derivative of M does not. In this case, the density diverges to infinity and the spacetime assumes a coordinate singularity. This singularity is called a *shell-crossing singularity*, which is when spherical shells from different coordinate χ values coincide to the same areal radius R .

3.4.1 Arbitrary functions of the LTB model

There are three arbitrary functions in the LTB solution, namely $t_{\text{bb}}(\chi)$, $M(\chi)$, and $E(\chi)$. We recall the Newtonian conservation of energy equation for a point mass m undergoing purely radial motion in a gravitational potential induced by a central mass M and compare it with (3.15a):

$$E = \frac{1}{2}m\dot{r}^2 - \frac{GMm}{r} \rightarrow \dot{r}^2 = \frac{2E}{m} + \frac{2GM}{r}. \quad (3.17)$$

Thus, we can interpret $M(\chi)$ as the total mass inside the shell of the coordinate χ and $E(\chi)$ as (half of) the total energy function per mass unit enclosed by the same shell [18]. This mass function is usually called the “Misner-Sharp” mass in the literature [10, 13, 28]. This is the total mass that generates the gravitational field of the spacetime. Note that this is different from the total integrated energy density $\tilde{M}$, or the sum of the masses of all the particles in the system. Suppose that there exists a matter-filled sphere from the center of symmetry $\chi = \chi_0$ (i.e $R(\tau, \chi_0) = 0$ for all τ) until some coordinate radius $\chi = \chi_s$. The total integrated energy density of this sphere is:

$$\begin{aligned}\tilde{M} &= \int \sqrt{-\det g} T_{00} dV \\ &= \int_0^{2\pi} \int_0^\pi \int_{\chi_0}^{\chi_s} \frac{\rho R^2 R' \sin^2 \theta}{\sqrt{1+2E}} dV \\ &= \int_{\chi_0}^{\chi_s} \frac{4\pi \rho R^2 R'}{\sqrt{1+2E}} d\chi.\end{aligned}\tag{3.18}$$

From (3.15b),

$$M = \int_{\chi_0}^{\chi_s} 4\pi \rho R^2 R' d\chi.\tag{3.19}$$

Hence, the difference between the total integrated energy density and Misner-Sharp mass depends entirely on the sign of the energy function. The system is said to be *bound* when $E < 0$, or when $\tilde{M} > M$. The difference $\tilde{M} - M$ is the binding energy of the system that keeps the configuration stable. When $E = 0$, the system is *marginally bound* and no energy was shed to form it nor is there excess energy.

Lastly, computing the 3-Ricci scalar on the hypersurfaces orthogonal to the fluid tangent vector gives us

$${}^3R = -\frac{2}{\chi^2} \frac{\partial}{\partial \chi} (\chi E),\tag{3.20}$$

which allows us to interpret E as also being related to the spatial curvature of the spacetime.

3.4.2 FLRW limit

As one would expect, the LTB class of solutions which is an inhomogeneous cosmological solution should reduce to the familiar homogeneous FLRW solution upon removing the dependence of the metric tensor components to the radial coordinate

function. This is in fact the case, and the covariant condition for the Friedmann limit of the LTB metric is $\partial_\chi \rho = 0$, or $\rho = \rho(\tau)$. Under the specific choice of geodesic coordinates, this translates to

$$t_{\text{bb}}(\chi) = \text{constant} \quad (3.21\text{a})$$

$$E(\chi) = -k\chi^2 \quad (3.21\text{b})$$

$$M(\chi) = \frac{4\pi\rho a^3}{3}\chi^3 \quad (3.21\text{c})$$

$$R(\tau, \chi) = a(\tau)\chi \quad (3.21\text{d})$$

where k and $a(\tau)$ is the curvature index and scale factor of the FLRW solution, respectively. Using the time translation invariance of the FLRW geometry, t_{bb} is usually set to zero. Also, despite $\rho(\tau)$ and $a(\tau)$ both being functions of time, the combination ρa^3 is constant from (3.10) upon plugging in $\omega = 0$. Substituting (3.21) to (3.14) and (3.15a) gives us the FLRW solution expressed in geodesic coordinates,

$$ds^2 = -d\tau^2 + a(\tau)^2 \left(\frac{d\chi^2}{1 - k\chi^2} + \chi^2 d\Omega^2 \right) \quad (3.22)$$

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi\rho}{3} - \frac{k}{a^2} \quad (3.23)$$

and (3.15b) becomes an identity.

3.5 Spacetime matching and junction conditions

Sometimes, it is in the interest of the researcher to patch together two (or more) space-time solutions which themselves constitutes a valid solution to the EFEs. Usually, this is done using spherically symmetric solutions which corresponds to a physical scenario where an interior spacetime is matched with an exterior one. To ensure smooth matching between the two solutions, we employ the *Darmois-Israel junction conditions*.

The spacetime domain $\mathcal{M}$ is split up into an interior $\mathcal{V}^-$ and exterior $\mathcal{V}^+$ region such that $\mathcal{M} = \mathcal{V}^- \cup \mathcal{V}^+$. These regions are joined along a boundary hypersurface, Σ . The two regions are endowed with different metric tensors g_{ab}^- and g_{ab}^+ and, in the most general case, are expressed in different coordinates $\{x_-^\mu\}$ and $\{x_+^\mu\}$. The first junction condition states that the induced metric on Σ from either side must be equal

$$h_{ab}^- = h_{ab}^+ \quad (3.24)$$

and expressed in the same coordinates, say $\{y^\gamma\}$ where $\gamma = 1, 2, 3$. The second junction condition requires that the extrinsic curvature also matches:

$$K_{ab}^- = K_{ab}^+. \quad (3.25)$$

Since the definition of the extrinsic curvature relies on defining a normal vector, a choice must be made on whether the normal vector points from $\mathcal{V}^-$ to $\mathcal{V}^+$ or vice versa. Either choice is valid and leads to a consistent solution. This condition is *optional*, in the sense that enforcing it results in having a vacuum 3-stress energy tensor S^{ab} . If the extrinsic curvature is not continuous, then $S^{ab} \neq 0$, and a thin shell of matter exists on the hypersurface, serving to balance the discontinuity in geometry.

To demonstrate how the junction conditions are applied, and to prepare for our discussion later on, we now introduce the classic *Oppenheimer-Snyder* solution.

3.5.1 Oppenheimer-Snyder collapsing spacetime

This solution, developed by J. Robert Oppenheimer and his student Hartland Snyder [1], was the first to model gravitational collapse into a black hole within general relativity. This paper is recognized as a foundational work that anticipated key features of modern gravitational collapse theory. They matched an interior $k = -1$ FLRW spacetime, sourced by a dust stress-energy tensor, with an exterior Schwarzschild vacuum. Essentially, what they are trying to model is a constant density star immersed in a vacuum background. The choice of an open geometry ($k = -1$) becomes clear upon examining (3.8a), which admits a turning point in the scale factor $a(\tau)$, ultimately leading to recollapse. We present a short pedagogical derivation here as adapted from [14] and [15]⁵.

The exterior region $\mathcal{V}^+$ is described by the Schwarzschild solution in its standard coordinates (3.3). The interior region $\mathcal{V}^-$, on the other hand, is described by the FLRW metric for a spatially closed universe, rewritten in comoving coordinates as

$$ds_-^2 = -d\tau^2 + a^2(\tau)(d\chi^2 + \sin^2 \chi d\Omega^2), \quad (3.26)$$

⁵The details of the derivation is inconsequential for the purposes of this thesis and only serves to show the difficulty of applying the junction conditions even for highly idealized spacetimes.

which is a coordinate transformed version of (3.6) for $k = -1$. Rewriting the Friedmann equation (3.8a), we obtain:

$$\dot{a}^2 + 1 = \frac{8\pi}{3}\rho a^2. \quad (3.27)$$

A turning point in the expansion occurs when $\dot{a} = 0$, corresponding to the moment when the scale factor reaches its maximum value $a_{\max}$. Setting $\dot{a} = 0$ in (3.27) gives:

$$\begin{aligned} 1 &= \frac{8\pi}{3}\rho a_{\max}^2 \\ \Rightarrow \rho a^3 &= \text{constant} = \frac{3}{8\pi}a_{\max}, \end{aligned} \quad (3.28)$$

where we know that $\rho a^3 = \text{constant}$ from (3.10). Using (3.28) we can rearrange Eq. (3.27) into the following form:

$$\dot{a}^2 + 1 = \frac{a_{\max}}{a} \quad (3.29)$$

which is an ordinary differential equation that admits the standard parametric solution:

$$a(\eta) = \frac{1}{2}a_{\max}(1 + \cos \eta), \quad \tau(\eta) = \frac{1}{2}a_{\max}(\eta + \sin \eta). \quad (3.30)$$

At the endpoints of the evolution:

$$\begin{cases} \eta = 0 : & a = a_{\max}, \tau = 0, \\ \eta = \pi : & a = 0, \tau = \frac{\pi}{2}a_{\max}, \end{cases}$$

we see that the model describes a collapse from rest to a final singularity over a proper time duration of

$$\tau_{\text{collapse}} = \frac{\pi}{2}a_{\max}. \quad (3.31)$$

We now specify the hypersurface Σ that joins the two regions. We aim to match the induced metric on both sides:

$$ds_{\Sigma}^2 = h_{ij} dy^i dy^j, \quad (3.32)$$

and impose $h_{ab}^- = h_{ab}^+$. From the interior FLRW region $\mathcal{V}^-$, we choose the hypersurface $\chi = \chi_0$, which selects a 2-sphere at a fixed comoving coordinate radius. This 2-sphere has an areal radius $r = a(\tau) \sin \chi_0$, and thus the induced metric on Σ in coordinates $y^\gamma = (\tau, \theta, \phi)$ becomes:

$$ds_{\Sigma}^2 = -d\tau^2 + a^2(\tau) \sin^2 \chi_0 d\Omega^2. \quad (3.33)$$

In the Schwarzschild exterior $\mathcal{V}^+$, we parametrize the matching surface by functions $t = T(\lambda)$, $r = R(\lambda)$, with affine parameter λ . The tangent vectors satisfy $dt = \dot{T} d\lambda$, $dr = \dot{R} d\lambda$, so that the induced metric on Σ in coordinates $y^\gamma = (\lambda, \theta, \phi)$ is:

$$ds_{\Sigma}^2 = -(F\dot{T}^2 - F^{-1}\dot{R}^2)d\lambda^2 + R^2(\lambda)d\Omega^2 \quad (3.34)$$

where $F(R) = 1 - \frac{2M_s}{R(\lambda)}$. Due to spherical symmetry, we naturally identify the angular coordinates on both sides, i.e., $(\theta, \phi)_{\text{Schw.}} = (\theta, \phi)_{\text{FLRW}}$. We also identify the affine parameter λ with the proper time τ of comoving observers on the boundary. Thus, equating (3.33) and (3.34) yields the junction conditions:

$$R(\tau) = a(\tau) \sin \chi_0 \quad F\dot{T}^2 - F^{-1}\dot{R}^2 = 1 \quad (3.35)$$

The first equation gives the trajectory $R(\tau)$ of the boundary, while the second can be integrated to determine the Schwarzschild time coordinate $T(\tau)$. Solving for $\dot{T}$, we find:

$$\dot{T} = \frac{1}{F(R)} \sqrt{\dot{R}^2 + F(R)} \equiv \frac{\beta(R, \dot{R})}{F(R)}. \quad (3.36)$$

As the parameter η evolves from 0 to π , the scale factor $a(\tau)$ evolves from a_{max} to zero, implying that $R(\tau) = a(\tau) \sin \chi_0$ also collapses from its maximum to zero. The matching hypersurface Σ was explicitly chosen to ensure that the induced metrics from both sides agree. This requirement constrains the motion of the boundary, effectively determining a unique trajectory $R(\tau)$ and time function $T(\tau)$. Further imposing the continuity of the extrinsic curvature and choosing the normal vector to point from $\mathcal{V}^-$ to $\mathcal{V}^+$, it can be shown that

$$M_s = \frac{4\pi\rho}{3} R^3 \quad (3.37)$$

Substituting to (3.31), we find that $\tau_{\text{collapse}} = \pi(R^3/8M_s)^{1/2} \sim \rho^{-1/2}$. As is evident here, spacetime matching is a lengthy process and requires careful foresight regarding the choice of the hypersurface Σ . A more economical, though less general, approach to spacetime matching involves finding two spacetimes that can be expressed in the same coordinate system and for which the metric tensor assigns the same physical

meaning to those coordinates in each region. In such cases, one can simply impose boundary conditions on the metric functions to smoothly transition from one solution to the other. This strategy was employed by Lasky and Lun [10] to construct a $3+1$ setup that we will use for our numerical simulations.

Chapter 4

Perfect fluid collapse

Using the ADM formalism introduced in Sec. 2.5, Lasky and Lun [10] derived a generic spherically symmetric spacetime and reduced field equations describing a perfect fluid. The discussion for this chapter is essentially rederiving their result to setup our analysis later on. The line element of the spacetime is given by

$$ds^2 = -\alpha^2 dt^2 + \frac{(\beta dt + dr)^2}{1 + E} + r^2 d\Omega^2, \quad (4.1)$$

where $\alpha(t, r)$ is the lapse function, $\beta(t, r)$ is the radial component of the shift vector $\beta^i = \beta(\partial_r)^i$, and $E(t, r)$ is a metric function to be determined. To maintain the Lorentzian signature of the metric tensor, we impose a lower bound of $E > -1$. Our coordinates adopt a Painlevé-Gullstrand type time slicing, where $t = \text{constant}$ hypersurfaces represent local simultaneous surfaces. We match this with a perfect fluid stress energy tensor

$$T^{ab} = (\rho + P)n^a n^b + P g^{ab} \quad (4.2)$$

where $\rho(t, r)$ is the energy density, $P(t, r)$ is the isotropic pressure¹, and n^a is the vector field tangent to the fluid. We demand this vector field to be timelike, normalized, and satisfy Frobenius' theorem such that it is hypersurface orthogonal. We choose the normal one-form $n_a = -\alpha \nabla_a t$, whose orthogonal hypersurfaces Σ_t are spatial ($t = \text{constant}$). Explicitly writing out its vector components, we have

$$n^a = \frac{1}{\alpha}(\partial_t)^a - \frac{\beta}{\alpha}(\partial_r)^a. \quad (4.3)$$

¹Since we want the final solution to describe an inhomogeneous spacetime, we allow the matter fields to vary with respect to the radial coordinate.

A simple calculation will show that

$$n^b \nabla_b n^a = \frac{1+E}{\alpha} \frac{\partial \alpha}{\partial r} (\partial_r)^a, \quad (4.4)$$

implying that the fluid is not in free fall. We project the conservation of the stress energy tensor $\nabla_a T^{ab} = 0$ along Σ_t which then provides us with another independent equation that belongs to the spatial hypersurfaces that foliate the spacetime.

$$\frac{1}{\alpha} \frac{\partial \alpha}{\partial r} = - \frac{1}{\rho + P} \frac{\partial P}{\partial r} \quad (4.5)$$

Substituting (4.5) back to (4.4) allows us to recover the fact that the presence of a pressure gradient prevents the free fall of the fluid. From inspecting the 4 remaining ADM equations, we find a particular combination of the metric functions will allow us to define the following scalar function

$$\frac{\partial M}{\partial r} := \frac{\partial}{\partial r} \left[r^2 (1+E) \frac{\partial}{\partial r} \ln \alpha + r^2 \mathcal{L}_n \frac{\beta}{\alpha} - 4\pi P r^3 \right] = 4\pi \rho r^2, \quad (4.6)$$

where $\mathcal{L}_n$ is the Lie derivative along the fluid tangent vector. Evaluating this function at $r = 0$ reveals that it vanishes, and integrating this up to some finite coordinate radius gives an equivalent expression to the integrated density of a sphere with the same radius in flat space. Hence, we adopt this as a natural “mass” function. Since this particular combination of metric functions appears throughout the ADM system, it is convenient to adopt this new mass function as another metric function.

Further simplification of the ADM equations will provide us with the following relation between the lapse and the shift:

$$\left(\frac{\beta}{\alpha} \right)^2 = \frac{2M}{r} + E, \quad (4.7)$$

which will allow us to do away with one of them. Because of (4.5), there is a fixed relation between the pressure and lapse function which necessarily ties it to the matter fields. Hence, we eliminate all explicit dependence on the shift vector by choosing

$$\beta = \alpha \sqrt{\frac{2M}{r} + E}. \quad (4.8)$$

Note that this choice is not unique. We may as well choose the negative root which is itself constitute a valid solution to the EFEs. Upon examining the ADM

line element and understanding the role of the shift vector, we understand that this choice is equivalent to choosing a collapsing spacetime as it dictates that the points on each spatial slice will be embedded into “smaller” coordinate values. The negative square root choice then is the time reversed direction and represents an expanding cosmological type solution. The implications of this choice will limit our choice of external universe later on and will become relevant in the discussion of Chapter 6.

After simplifying the ADM equations, we obtain two independent equations that live on the spatial hypersurfaces orthogonal to the fluid’s tangent vector which are a coupled system of nonlinear hyperbolic equations governing the metric functions $M(t, r)$ and $E(t, r)$.

$$\frac{\partial M}{\partial t} = \alpha \sqrt{\frac{2M}{r} + E} \left(\frac{\partial M}{\partial r} + 4\pi P r^2 \right) \quad (4.9a)$$

$$\frac{\partial E}{\partial t} = \alpha \sqrt{\frac{2M}{r} + E} \left(\frac{\partial E}{\partial r} + 2 \frac{1 + E}{\rho + P} \frac{\partial P}{\partial r} \right) \quad (4.9b)$$

This, along with (4.5), leaves us with three equations for four unknowns, which will be closed by providing an equation of state. It is crucial to highlight that these equations are simply the Lie derivative of the metric functions along the fluid tangent vector:

$$\mathcal{L}_n M = 4\pi P r^2 \sqrt{\frac{2M}{r} + E} \quad (4.10a)$$

$$\mathcal{L}_n E = 2 \frac{1 + E}{\rho + P} \sqrt{\frac{2M}{r} + E} \frac{\partial P}{\partial r} \quad (4.10b)$$

An interesting thing to point out in this solution is that we end up with field equations that are first order PDEs, as opposed to second order which is usually the case for EFEs. This is a peculiarity that could be pointed as one of the main results of Lasky and Lun, as they have eliminated second order derivatives by carefully massaging the ADM system of equations into the following form.

At this point, we have strictly treated the field equations under the ADM formalism without any reference to the inhomogeneous LTB solution. We now show that this $3 + 1$ solution is in fact equivalent to the LTB spacetime and there exists a coordinate transformation from the PG chart (t, r) of (4.1) to the geodesic chart (τ, χ) of (3.14). We can see that the PG chart is non-diagonal, which suggests that

$$\beta dt + dr = \frac{\partial R}{\partial \chi} d\chi \quad (4.11)$$

to obtain a diagonal line element. Explicitly writing out the total differentials of the coordinate functions provides us the following two equations:

$$\beta \frac{\partial t}{\partial \tau} + \frac{\partial r}{\partial \tau} = 0, \quad (4.12a)$$

$$\beta \frac{\partial t}{\partial \chi} + \frac{\partial r}{\partial \chi} = \frac{\partial R}{\partial \chi}. \quad (4.12b)$$

Since both r and R are the areal radius of their respective charts, we set $r(\tau, \chi) \equiv R(\tau, \chi)$ and (4.12b) implies that $t = t(\tau)$. Choosing the time slicing $t(\tau) \equiv \tau$ will give us

$$\beta = -\frac{\partial R}{\partial \tau} \quad (4.13)$$

from (4.12a). The line element then becomes

$$ds^2 = -\alpha(\tau, \chi)^2 d\tau^2 + \frac{1}{1 + E(\tau, \chi)} \left(\frac{\partial R}{\partial \chi} \right)^2 d\chi^2 + R(\tau, \chi)^2 d\Omega^2, \quad (4.14)$$

and substituting our calculations to (4.6) and (4.7) gives us

$$\left(\frac{\partial R}{\partial \tau} \right)^2 = \alpha^2 \left(\frac{2M(\tau, \chi)}{R} + E(\tau, \chi) \right), \quad (4.15a)$$

$$\frac{\partial M}{\partial \chi} = 4\pi \rho R^2 \frac{\partial R}{\partial \chi} \quad (4.15b)$$

For $n_a = -\alpha \nabla_a \tau$, its vector component is now only $n^a = (\partial_\tau)^a / \alpha$ and (4.10) now reads

$$\frac{\partial M}{\partial \tau} = 4\pi P R^2 \alpha \sqrt{\frac{2M}{R} + E}, \quad (4.16a)$$

$$\frac{\partial R}{\partial \chi} \frac{\partial E}{\partial \tau} = 2 \frac{1 + E}{\rho + P} \frac{\partial P}{\partial \chi} \alpha \sqrt{\frac{2M}{R} + E}, \quad (4.16b)$$

noting that $\partial_R = (\partial_\chi / \partial R) \partial_\chi$. Lastly, the orthogonal projection of $\nabla_a T^{ab} = 0$ gives

$$\frac{1}{\alpha} \frac{\partial \alpha}{\partial \chi} = -\frac{1}{\rho + P} \frac{\partial P}{\partial \chi}. \quad (4.17)$$

By restricting our choice of perfect fluid to strictly dust ($P = 0$), the equation above gives $\alpha = \alpha(\tau)$ and hence the lapse function can be rescaled to unity from (2.31). Moreover, (4.16) then implies that both scalar functions are strictly functions

of χ . It is then evident that the derived 3+1 solution reduces to the LTB system of equations² when considering a dust matter source, and the scalars M and E are the Misner-Sharp mass and energy function of the LTB solution. It is crucial to highlight that the “new form” of the energy density definition (4.6) in these coordinates would make one naively think that shell-crossing singularities are no longer an issue. This is in fact *not* the case, and this coordinate singularity still pops up in this formalism through multi-valued solutions of the mass function. This will be expounded more later on.

Since our solution belongs to the family of LTB solutions, it shares the coordinate invariant condition that imposing $\rho = \rho(t)$ reduces the EFEs into the Friedmann equations. The mass function then becomes $M(t, r) = 4\pi\rho(t)r^3/3$ due to (4.6). Supposing that the entire spacetime is sourced by a global cosmic fluid described by the equation of state $P = P(\rho)$, (4.5) dictates that the lapse function becomes strictly a function of time again. The field equations can then be rewritten as a coupled set of ordinary differential equations:

$$\frac{d\rho}{dt} = 3\sqrt{\frac{8\pi\rho}{3} + \epsilon}(\rho + P) \quad (4.18a)$$

$$\frac{d\epsilon}{dt} = 2\epsilon\sqrt{\frac{8\pi\rho}{3} + \epsilon} \quad (4.18b)$$

where $E(t, r) = \epsilon(t)r^2$. Note that this form of E is necessary to maintain consistency of functional dependence within the equations. The line element then reads:

$$ds^2 = -dt^2 + \frac{1}{1 + \epsilon(t)r^2} \left(\sqrt{\frac{8\pi\rho}{3} + \epsilon} dt + dr \right)^2 + r^2 d\Omega^2. \quad (4.19)$$

Further imposing that the equation of state is linear ($P = \omega\rho$) will give an exact solution to (4.18) given by:

$$\rho \propto \epsilon^{3(1+\omega)/2} \quad (4.20)$$

Comparing this with (3.12) (which we rewrite here for convenience),

$$ds^2 = -dt^2 + \frac{1}{1 - kr^2/a^2} \left(\frac{\dot{a}}{a} r dt + dr \right)^2 + r^2 d\Omega^2, \quad (4.21)$$

²Refer to Section 3.4.

we note the explicit similarities between the two line elements. After coupling this metric with the same linear equation of state, a quick calculation of the Einstein equations in these coordinates will show that the Friedman equations will have the same form and have the solution

$$\rho \propto a^{-3(1+\omega)} \quad (4.22)$$

It is then evident that these two solutions are the same by choosing $\epsilon(t) = -k/a^2$, making use of the positive root of (3.8a) to equate the radial shift vector components.

4.1 Well-posedness of the system of equations

In theory, the manner in which the field equations (4.9) are derived makes it conducive to a Cauchy problem formulation. One could simply supply the initial data $M_0(r)$ and $E_0(r)$ on an initial hypersurface and the equations will handle the entire evolution of the scalar functions across the entire manifold. If such a solution does exist and is found, it must be unique (i.e. single-valued everywhere), and depend continuously on the initial data (to preserve causality) [29]. The metric function solutions, whether obtained analytically or numerically, then allows us to fully determine the entire history and future of the spacetime manifold.

However, there are latent issues with this formalism. Mathematically, the field equations are a system of coupled first-order hyperbolic advection equations for two scalar functions M and E that are sourced by the pressure function. Rewriting these equations, we have

$$\frac{\partial \vec{U}}{\partial t} + \mathbb{A} \frac{\partial \vec{U}}{\partial r} = \mathbb{S} \quad (4.23)$$

where $\mathbb{A} = -\alpha \sqrt{E + \frac{2M}{r}} \mathbb{I} = -v \mathbb{I}$ is the coefficient matrix and $\mathbb{S} = v \left(4\pi P r^2 \quad 2 \frac{1+E}{\rho+P} \frac{\partial P}{\partial r} \right)^T$ is the source matrix. An additional condition to guarantee well-posedness in systems of hyperbolic equations is *strict hyperbolicity*. A set of hyperbolic PDEs is strictly hyperbolic if its coefficient matrix can be diagonalized, its eigenvalues are real and distinct, and its eigenvectors form a basis set [30, 31]. Since $\mathbb{A} \propto \mathbb{I}$, it is trivially diagonalizable and any choice of two linearly independent eigenvectors will form a basis set. However, its eigenvalues are degenerate and must satisfy $E + 2M/r \geq 0$ for all t and r . Imposing that the perfect fluid stress-energy tensor follows at the bare minimum the weak energy condition, we are guaranteed that the Misner-Sharp mass

is strictly positive. We are then limited on the form of E since the only condition it must follow is $E > -1$. Therefore, it could possibly be the case that E evolves in a way such that the characteristic velocity becomes complex-valued at some point in the spacetime manifold, losing the hyperbolicity condition. A more comprehensive discussion of the hyperbolicity condition for systems of advection equations can be found in Chapter 16 of [31]. Overall, we are not guaranteed that the system (4.9) is strictly hyperbolic, and hence well-posed.

The most appropriate and commonly used method for solving these types of partial differential equations is the method of characteristics [30, 32, 33]. Both scalars are being transported by the characteristic velocity

$$v(r, M, E) = \alpha \sqrt{\frac{2M}{r} + E} \quad (4.24)$$

From (4.5), the prescription of a global equation of state, and (4.6) the lapse function is a function of M . Due to the highly nonlinear form of v and its dependence on the scalar functions, it is highly likely that the characteristic curves of both PDEs will intersect and form caustics due to differences in velocity of adjacent curves. In the event of caustic or shock formation, the metric functions become multivalued, which signals a breakdown in the coordinates of the spacetime manifold. Anyhow, the scalar fields themselves can be evolved past shock discontinuities using weak solutions. In particular, advection equations that can be written in conservation form (or in Fick's laws form) are preferred due to the Lax-Wendroff theorem [34]. For the advective scalar field $\psi(x^\mu)$ described by the inhomogeneous partial differential equation

$$\frac{\partial \psi}{\partial t} + v(x^\mu, \psi) \frac{\partial \psi}{\partial r} = F(x^\mu), \quad (4.25)$$

its conservation form is written as

$$\frac{\partial \psi}{\partial t} + \frac{\partial j}{\partial r} = F(x^\mu), \quad (4.26)$$

where $j(x^\mu, \psi)$ is its flux field. If one could find an appropriate form for the flux function of the scalar ψ , the Lax-Wendroff theorem guarantees the stability and convergence of valid numerical schemes to a weak solution [31, 35]. This poses further problems for the system of equations. Since the coupled scalar fields share the same characteristic velocity, it is impossible to find an appropriate coordinate transformation that will allow us to write a flux function for both M and E . Hence,

we are not guaranteed that whatever numerical scheme we impose will be stable and convergent to a weak solution.

Furthermore, we are faced with the task of providing an appropriate initial function $E_0(r)$. In the original derivation of the LTB solution, the energy function $E(\chi)$ is a completely arbitrary function of the radial coordinate χ that comes up as an integration “constant” in the field equations. Because of this, one is free to choose any functional form of $E(\chi)$ and it will be a completely valid member of the LTB family of solutions. A particularly useful form of this arbitrary function is $E \sim k\chi^2$, where k becomes the curvature of the equivalent FLRW spacetime in the appropriate limit ($\rho = \rho(t)$). Because of this, the most common approach done in the literature is to choose an energy function that asymptotically approaches a quadratic dependence for large χ to obtain an exact spacetime solution whose interior is LTB and whose exterior is FLRW. Another typical strategy is to obtain an exact spacetime that produces the spherical top-hat collapse model ([18, 24] and references therein). In any case, due to its arbitrariness and the lack of concrete interpretation on this function, it is not immediately obvious how one could provide a physically meaningful initial energy function. In fact, the original authors themselves mistakenly prescribed readers to use the evolution equation for M to provide an initial $E_0(r)$, which invalidates the simultaneous evolution of both metric functions (Equation 36 of [10]).

Because of all of these complications, no analytic and numerical solution has been found for the full system of equations and it remains to be an open problem [28, 36], with the authors of the original paper (Lasky and Lun) publishing their last paper about this topic in 2009 [37]. At this point, we would like to emphasize that a significant portion of the author’s research time was used to obtain a solution to the full system of PDEs, but to no avail.

More recently, examinations of this spacetime under the effective loop quantum gravity formalism has shown that the scalar fields M and E might not be the best choice of fundamental fields to consider as metric functions, and that a better choice might be to use a component of the Ashtekar-Barbero connection ([38] and references therein). For the purposes of this thesis, we disregard this and restrict ourselves to classical general relativity.

4.2 Marginally bound case

In order to progress, we restrict our analysis to the marginally bound LTB case and set $E = 0$ everywhere. This accounts to having $k = 0$ in the FLRW limit. The 3-Ricci tensor on the orthogonal hypersurfaces to n^a vanishes, implying spatial flatness. Substituting $E = 0$ onto 4.9b shows that this is a valid solution given that the pressure of the matter source is homogeneous and strictly a function of time ($P = P(t)$). The authors of [10] argued that imposing a dust equation of state would mean that the density must necessarily be a function of time only everywhere, implying that setting $E = 0$ is tantamount to simplifying the LTB solution into an FLRW. However, this is only true when imposing a *global* equation of state. One could impose *local* equations of state, such that $P(t) = P(\rho_b)$ for a local liquid with energy density $\rho_b = \rho_b(t)$ which would then be a genuine FLRW region sourced by a cosmic fluid. An energy density at another region can be defined such that $\rho_{\text{int}} = \rho_{\text{int}}(t, r)$ and the inhomogeneity of the spacetime is maintained. However, to preserve the strict time-dependence of the pressure of this local fluid, the only valid form is $P = 0$; i.e. dust. Thus, we are allowed to smoothly match a marginally bound ($E = 0$) LTB interior and spatially flat ($k = 0$) exterior FLRW region with one coordinate patch by studying the perfect fluid mixture given by

$$\rho(t, r) = \rho_{\text{int}}(t, r) + \rho_b(t) \quad P_{\text{int}}(t) = 0 \quad \text{and} \quad P_b(t) = P(\rho_b) \quad (4.27)$$

Since the pressure is now homogeneous everywhere, the lapse is again removable with coordinate freedom. The reduced line element then is

$$ds^2 = -dt^2 + \left(\sqrt{\frac{2M}{r}} dt + dr \right)^2 + r^2 d\Omega^2, \quad (4.28)$$

with the remaining field equation being

$$\frac{\partial M}{\partial t} - \sqrt{\frac{2M}{r}} \frac{\partial M}{\partial r} = \sqrt{2M} \, 4\pi P(t) r^{3/2}. \quad (4.29)$$

Recall that this single remaining metric function is coupled with the density via (4.6) and the pressure after imposing our specific equation of state. Hence, the full manifold is entirely determined by the matter source. We would like to emphasize here that this choice inevitably limits our formalism to not admit radial pressure gradients in

the matter source. It is for this reason that we only consider the collapse of dust for our interior matter. Because of the absence of pressure, there is no mechanism in our model that prevents localized matter from repelling each other. As such, we should expect that the gravitational collapse occurring in the spacetime is inevitable once it has begun.

One can notice that the form of (4.28) is exactly the form of (3.4) but with a scalar function $M(t, r)$. As such, it is no surprise that a trivial solution to (4.29) is $M = \text{const.}$ with vanishing pressure. By letting the energy density vanish at some finite boundary r_b , we can obtain a constant mass function for $r > r_b$ by utilizing (4.6). Hence, an appropriate choice of energy density function will allow us to smoothly match an interior LTB spacetime with an exterior Schwarzschild using the same coordinate patch, avoiding the need for Darmois-Israel junction conditions across a hypersurface boundary [10].

Chapter 5

Vacuum background

As a starting point, we consider the simplest case for our spacetime which is when we consider a vanishing homogeneous pressure and assume a stationary vacuum exterior to the inhomogeneous dust. The field equation (4.29) reduces to

$$\frac{\partial M}{\partial t} - \sqrt{\frac{2M}{r}} \frac{\partial M}{\partial r} = 0. \quad (5.1)$$

This partial differential equation for $M(t, r)$ is exactly solvable and has already been found [35] which we rederive here. The key insight of the method of characteristics is that the scalar we are solving for, in this case M , remains constant along its characteristic curves¹. In essence, we reduce this nonlinear partial differential equation into a system of ordinary differential equations [32, 33]. Providing an initial function $M_0(r) := M(0, r)$ will allow us to compute the conserved value of M at each point in space that will then be transmitted along its characteristics.

We perform a change of variables from (t, r, M) to (τ, x, μ) and impose that the parameter τ is increasing along the direction of the characteristic curves of M . The auxiliary function $\mu(\tau, x)$ is defined such that $M(t, r) = \mu(\tau(t, r), x(t, r))$. Therefore, finding a functional form for μ is equivalent to obtaining a solution to the equation. The system of ODEs then is:

$$\frac{\partial t}{\partial \tau} = 1 \quad t(0, x) = 0 \quad (5.2a)$$

$$\frac{\partial r}{\partial \tau} = -\sqrt{\frac{2\mu}{r}} \quad r(0, x) = x \quad (5.2b)$$

$$\frac{\partial \mu}{\partial \tau} = 0 \quad \mu(0, x) = M_0(x) \quad (5.2c)$$

¹We will find later on that this fact translates into the timelike geodesics of the spacetime carrying its own conserved mass.

From (5.2a) and (5.2c), we readily see that $t(\tau, x) = \tau$ and $\mu(\tau, x) = M_0(x)$. Solving the remaining equation gives us $r^{3/2} = x^{3/2} - \frac{3}{2}\tau\sqrt{2\mu(\tau, x)} = x^{3/2} - \frac{3}{2}\tau\sqrt{2M_0(x)}$. Thus, the particular solution to (5.1) after imposing an initial condition is

$$M(t, r) = M_0 \left(\left(r^{3/2} + \frac{3}{2}t\sqrt{2M(t, r)} \right)^{2/3} \right). \quad (5.3)$$

Since this closed form solution is implicit, we obtain approximate solutions for $M(t, r)$ using root-finding algorithms. We do this by defining some computational region $\{(t, r) \mid t \in [t_{\min}, t_{\max}], r \in [r_{\min}, r_{\max}]\}$ and discretizing it such that we obtain $N \times N$ pairs of grid points (t_i, r_j) . We then input each grid point values into (5.3) and root-find for values of M that will equate the left and right hand sides of the relation². Hence, we expect that the solution for $M(t, r)$ is some 3D surface in (t, r) . These discrete points can then be interpolated to obtain a smooth and continuous solution. Due to its implicit nature, it is possible that the mass could be multivalued at some points. Since $M(t, r)$ is a metric function, its multivaluedness in some (t, r) region/s signals a breakdown in the coordinates. Thus, we check where this occurs by evaluating the Jacobian.

The Jacobian matrix $\mathcal{J}$ of the transformation from $(\tau, x) \rightarrow (t, r)$ is given by

$$\mathcal{J} = \begin{pmatrix} \frac{\partial t}{\partial \tau} & \frac{\partial t}{\partial x} \\ \frac{\partial r}{\partial \tau} & \frac{\partial r}{\partial x} \end{pmatrix}. \quad (5.4)$$

The solution (5.3) will become multi-valued when its characteristic curves cross or when the determinant of this Jacobian matrix vanishes; i.e. when $\partial r / \partial x = 0$. It has been shown [35] that this corresponds to a condition on the associated tidal forces of the spacetime $\Phi(t, r)$. This condition is

$$\Phi(0, r) = -\frac{r}{3} \frac{\partial}{\partial r} \left(\frac{M_0}{r^3} \right) < 0 \quad (5.5)$$

for some $r \in [0, +\infty)$. The converse is also true; if $\Phi(0, r) \geq 0$ for all r , then M will remain single-valued throughout. For our purposes, we want to translate this into a condition on $\rho_0(r)$.

²I would like to credit Sean Julian Fortuna for suggesting this method of obtaining solutions for $M(t, r)$.

Performing the differential operator, we find that M will be a single-valued function everywhere if $M_0(r) \geq rM'_0(r)$. Using (4.6), this translates into the following condition on ρ_0 :

$$\rho_0(r)r^3 \leq \int_0^r \rho_0(x)x^2 dx \quad (5.6)$$

Suppose that $\rho'_0(r) > 0$ at a point $r = r_1$, which means that $\rho_0(r)$ is increasing with r in some neighborhood around r_1 . Thus, $\rho_0(x) < \rho_0(r_1)$ for all $x \in [0, r_1)$. It thus follows that

$$\begin{aligned} \int_0^{r_1} \rho_0(x)x^2 dx &< \int_0^{r_1} \rho_0(r_1)x^2 dx \\ 3 \int_0^{r_1} \rho_0(x)x^2 dx &< \rho_0(r_1)r_1^3, \end{aligned}$$

taking note that $\rho_0(r_1)$ is a constant. This is clearly a contradiction of (5.6). Hence, $\rho'_0(r)$ has to be less than zero for $r \in [0, +\infty)$. Therefore, *we restrict our choices of ρ_0 to monotonically decreasing functions* to ensure that our numerical simulations will be valid throughout.

From (5.3), we find that the characteristic curves of this PDE is

$$r(t; r_0) = \left(r_0^{3/2} - \frac{3}{2}t\sqrt{2M_0(r_0)} \right)^{2/3} \quad (5.7)$$

where $r(t=0) := r_0$ is a parameter that distinguishes characteristics and $M_0(r_0) = M(0, r(0)) = M(t, r(t))$ is the constant value of M along the characteristic. To obtain a more geometrical understanding of (5.1), we recall that the left hand side is simply the Lie derivative of M along the fluid's tangent vector. Since $\mathcal{L}_n M = 0$ the mass is conserved along the worldlines whose tangent vector is n^a . From (4.4), we know that

$$n^a = (\partial_t)^a - \sqrt{\frac{2M}{r}}(\partial_r)^a \quad (5.8)$$

is now a geodesic. Hence, $n^a := dx^a/d\lambda$ where λ is the affine parameter along the geodesic. It is then evident that t parametrizes the radial coordinate and the radial worldlines $x^a(\lambda)$ are the solutions to the ODE

$$\frac{dr}{dt} = -\sqrt{\frac{2M(t, r)}{r}}. \quad (5.9)$$

We make use of the fact that M is conserved along the integral curves of n^a and

substitute $M(t, r(t)) = M(0, r(0)) := M_0(r_0)$ where $r(t=0) := r_0$ and $M_0(r_0)$ is the conserved mass. Equation (5.9) now has the solution

$$r(t; r_0) = \left(r_0^{3/2} - \frac{3}{2}t\sqrt{2M_0(r_0)} \right)^{2/3}. \quad (5.10)$$

Here, r_0 is a continuous parameter that labels each geodesic and t is the affine parameter (proper time) along the fluid flow as expected. We now see that the characteristic curves of the PDE is exactly just the congruence of timelike geodesics of the spacetime, each of whom carry their own conserved mass. It follows that freely falling observers who are comoving with the fluid measure a proper time between two events that is equal to the difference in coordinate time between said two events. This key property of the timelike observers will be used extensively for our analysis later.

We can invert this expression to instead express the proper time measured by the observer originating at r_0 to arrive at an arbitrary r :

$$t(r; r_0) = \frac{2}{3\sqrt{2M_0(r_0)}}(r_0^{3/2} - r^{3/2}). \quad (5.11)$$

Taking the limit³ of this as $r_0 \rightarrow 0$ and evaluated at $r = 0$, we find an important value that we will label as t_{crit}

$$t_{\text{crit}} = \lim_{r_0 \rightarrow 0} \frac{2r_0^{3/2}}{3\sqrt{2M_0(r_0)}} = \frac{1}{\sqrt{6\pi\rho_0(0)}} = \frac{1}{\sqrt{6\pi\rho_c}}, \quad (5.12)$$

where we used (4.6) to recast M_0 into ρ_0 . This result implies that observers comoving with dust shells whose initial position is extremely close to $r = 0$ will not immediately reach the center but will instead hover for some minimum time. Due to this “free-fall lag”, neighboring characteristics near the center ($r_0 \ll 1$) will cluster at some time $t > t_{\text{crit}}$. Since this model admits no mechanism that prevents the further collapse of shells towards the center (due to the absence of pressure), there will always be a singularity at $r = 0$ due to the focusing of the freely falling dust shells. An important point to note here is that this result can be obtained before specifying any particular form of initial density or mass functions, only requiring the value of $\rho_0(r)$ at $r = 0$. Due to its generality, this remark applies to all spacetimes

³For some choices of M_0 , taking the limit of $t(0; r_0)$ as $r_0 \rightarrow 0$ sometimes leads to indeterminate forms that cannot be removed by L’Hopital’s rule. In those cases, taking the Taylor series expansion around $r_0 = 0$ instead is sufficient and will provide the same result.

described by (4.28) and (5.1). Because $t_{\text{crit}} \sim \rho_c^{-1/2}$, we find that more compact stars collapse faster than sparse ones.

Since we expect that a singularity will form in our system, it is only natural to also expect that our matter forms a black hole to conceal the singularity. Thus, we must track the presence and formation of apparent horizons. By utilizing the Euler-Lagrange equations, we can compute for the radial null geodesics using the line element. The tangent vector of the radial null geodesics is $k^a = \partial_t + \left(\pm 1 - \sqrt{2M/r}\right) \partial_r$, with the positive/negative root corresponding to the outgoing/ingoing null rays. Then, the congruence of null geodesics are the solutions to the ODE

$$\frac{dr}{dt} = \pm 1 - \sqrt{\frac{2M(t, r)}{r}}, \quad (5.13)$$

coupled with the condition $r(t_i) = r_j$ for the radial null geodesic originating from the event (t_i, r_j) . Unlike the timelike case, we can no longer expect the mass to be conserved along the null geodesics. Since we obtain M as a discrete set through root-finding, we turn it into an interpolating function so that it accepts inputs of (t, r) . Therefore, we obtain the integral curves of this ODE through full-fledged numerical integration.

The slope of the integral curves of the outgoing geodesics then admits a turning point when

$$r_{\text{AH}}(t) = 2M(t, r_{\text{AH}}(t)). \quad (5.14)$$

On this surface, initially outgoing null rays become stationary before infalling back into smaller r values. Therefore, this relation defines the location of the apparent horizon on the current time slice. Interestingly, solving for the vanishing of the expansion scalar $\Theta = \nabla_a k^a$ results in the exact same relation. Again, we employ root-finding schemes due to the implicit nature of the relation. Unlike the mass function, it is acceptable to obtain multivalued solutions for r_{AH} for a given time slice.

5.1 Oppenheimer-Snyder dust collapse

As a numerical test bed, we start by recreating the Oppenheimer-Snyder spacetime. Recalling that the physical system described in the OS spacetime is a constant density

star in a vacuum, we choose an initial energy density profile on the initial time slice and parametrize it as follows:

$$\rho_0(r) = \begin{cases} \rho_c & \text{if } r \leq r_b, \text{ and} \\ 0 & \text{if } r > r_b. \end{cases} \quad (5.15)$$

Inside the boundary region ($r < r_b$), the density is not a function of radius. Since we have $E = 0$ the region is exactly described by a flat ($k = 0$) FLRW, as opposed to the typical closed ($k = +1$) FLRW used in OS spacetimes. This usual choice is done such that the Friedmann equation (3.8a) for the scale factor $a(t)$ admits a turning point. Hence, there is a recollapse phase which is what “triggers” the collapse. This choice of energy density is not C^1 differentiable, but it is not issue because the metric function is the mass, which is continuous.

For our simulations, we shall use the the solar mass $M_\odot \approx 1.5 \text{ km}$ as the characteristic length scale. In geometrized units, the parameter ρ_c is in units of $M_\odot^{-2}$ while r_b is in units of $M_\odot$. The initial mass data would then be given by

$$M_0(r) = \begin{cases} \frac{4\pi\rho_c r^3}{3} & \text{if } r \leq r_b, \text{ and} \\ \frac{4\pi\rho_c r_b^3}{3} & \text{if } r > r_b, \end{cases} \quad (5.16)$$

which is visualized as the blue curve in Fig. 5.1. The value of M_0 at any $r \leq r_b$ denotes the star’s integrated or cumulative mass up until that point. Past $r = r_b$, the mass function becomes constant and the spacetime is exactly a Schwarzschild vacuum with mass $M_s = 4\pi\rho_c r_b^3/3$. As we will see later, this could either be the OS star’s mass or the effective mass of a black hole depending on the initial configuration. We denote the boundary of the star in the current time slice as the radius wherein the mass function plateaus and becomes a constant.

Evolving the Misner-Sharp mass according to the field equation solution (5.3) then provides us with a 3D surface plot for $M(t, r)$. For visualization, we have Fig. 5.1 which are snapshots $M(t_i, r)$ for the given choices of t_i . The radial mass profile evolves such that the boundary region between the star and the exterior vacuum continuously decreases up until some critical time t_{crit} . After this critical time the entire mass of the star will be deposited at $r = 0$, hinting at the possible collapse of the matter into a singularity.

An important point to note here is that the axes of the plot are scaled such that the initial radius of the star and its total mass will always be located at 1 on the

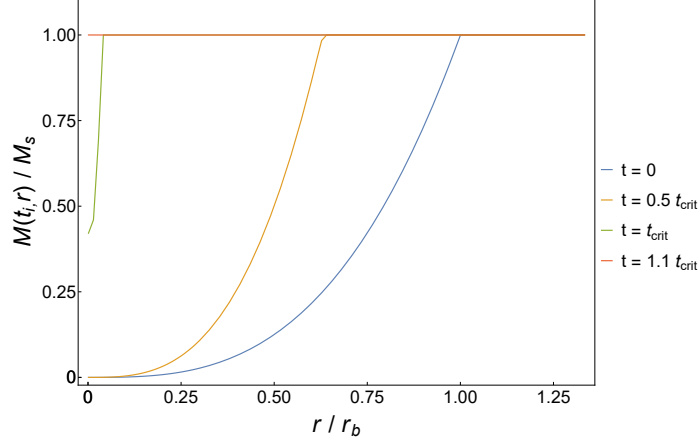


Figure 5.1: Mass evolution of the constant density matter.

respective axes. This was done because we find that the shape and evolution of the Misner-Sharp mass does not qualitatively depend on the input parameters ρ_c and r_b . The parameters merely act as scaling factors to the total value of M_s , but all choices will lead to the same qualitative result; the star surfaces starts off at $r = r_b$ and collapses into its center after some time $t_{\text{coll}} = t_{\text{crit}} \sim \rho_c^{-1/2}$ exactly like the Oppenheimer-Snyder solution.

In order to verify if what we are observing here is indeed matter collapsing into a singularity, we observe the formation of apparent horizons. We find that the behavior of apparent horizon formation largely depends on the choice of initial parameters. By observing the dependence of the existence of roots of (5.14) on the parameters, we find that there are two horizon formation scenarios that occur depending on the initial configuration of the homogeneous star:

1. Two initial AHs; the outer horizon is at the Schwarzschild radius ($r = 2M_s$) and the inner horizon gradually shrinks until it approaches $r = 0$ at $t = t_{\text{crit}}$.
2. No initial AHs, then one appears at $r = 2M_s$. An inner horizon then splits from this which then disappears at $t = t_{\text{crit}}$.

An intuitive way to think about this is that our choice of initial parameters determines if the star we constructed on the initial hypersurface is already dense enough to construct an apparent horizon or not. We can think of case 1 as the future scenario of case 2 after the appearance of its outer horizon. We reserve the discussion of the specific condition for each case later.

The formation of the apparent horizon with respect to the mass function evolution can be seen in Fig. 5.2⁴. We have chosen input parameters that belong in each of the two cases mentioned above. We can see more clearly that the entire star is already inside the apparent horizon in case 1 while case 2 describes the case where the star collapses into its Schwarzschild radius. It is evident that the entire mass of the star collapses into $r = 0$ at $t = t_{\text{crit}}$.

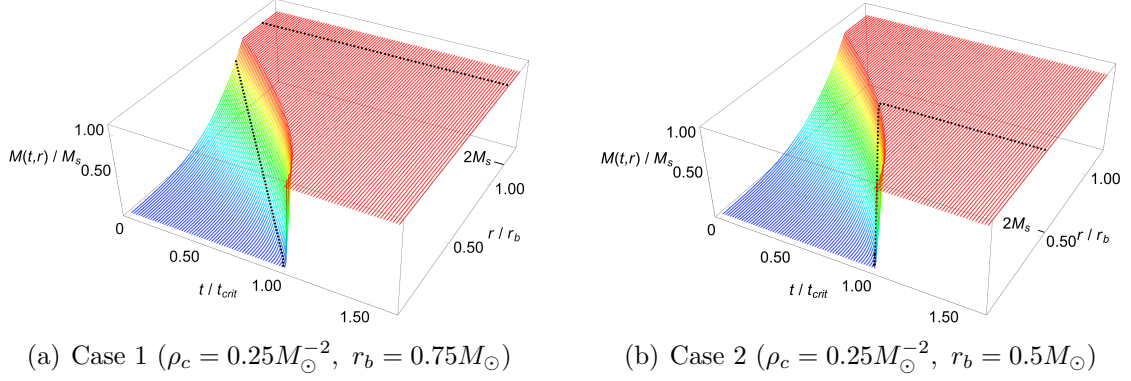


Figure 5.2: Time evolution of the Oppenheimer-Snyder mass function for two different input parameters. Color denotes relative values of mass at each event. The dashed lines show the apparent horizons.

This evolution of M is measured by the stationary observers of the spacetime, each whom individually report the value of M at their fixed radius. We can shift our perspective and instead view the collapsing star as a collection of concentric, infalling dust shells. This is done by following the worldlines of observers comoving with each shell of dust; the timelike geodesics which carry its own conserved mass. Explicitly, the worldline of the i^{th} dust shell situated at r_i on the initial hypersurface is given by⁵:

$$r_i(t) := r(t; r_i) = \begin{cases} r_i \left(1 - \frac{t}{t_{\text{crit}}}\right)^{2/3} & \text{if } r_i \leq r_b \text{ and} \\ r_i \left(1 - \frac{t}{t_{\text{crit}}} \frac{r_b^{3/2}}{r_i^{3/2}}\right)^{2/3} & \text{if } r_i > r_b \end{cases} \quad (5.17)$$

This highlights the simultaneous collapse of the dust shells; $r_i(t)$ will vanish at $t = t_{\text{crit}}$ regardless of r_i or where it originated from.

⁴We note that $M(t, r)$ consistently becomes multivalued at $r = 0$ past the critical time for all the cases that we considered, and hence we excise it from our analysis.

⁵Again, note the abuse of notation; $r_i(t)$ denotes the position of the i^{th} dust shell at some later time, while r_i is its initial position.

With this, we can define the deviation vector for the geodesics inside r_b (noting that $\lambda = t$):

$$\xi^a := \frac{\partial x^a(\lambda)}{\partial r_i} = \left(1 - \frac{t}{t_{\text{crit}}}\right)^{2/3} (\partial_r)^a \quad (5.18)$$

We can see that the dust shells are only varying along the radial direction (indicating spherical collapse) and that at $t = t_{\text{crit}}$ all geodesics will simultaneously cross, forming a caustic. Since these are also the characteristic curves, this crossing also signals that M is now multi-valued at $r = 0$. Using the geodesic deviation equation, we find that

$$R_{bcd}^a n^b n^c \xi^d = -\frac{2}{9} \frac{1}{t_{\text{crit}}^2} \left(1 - \frac{t}{t_{\text{crit}}}\right)^{-4/3} (\partial_r)^a, \quad (5.19)$$

which indicates that the tidal forces diverges to infinity at t_{crit} , which is strong evidence for the formation of a geometric singularity. The spacetime diagram of the OS collapse for each of the cases can be seen in Fig. 5.3. The simultaneous collapse of the dust shells as well as the free-fall lag of the central ones can be clearly seen in this diagram. We also see the contrast between the two parameter choices with regards to the star's Schwarzschild radius. In the first case, the matter is already inside $r = 2M_s$ in the initial time slice and thus is initialized as a black hole. For the second, the matter starts off as a star before collapsing into a black hole.

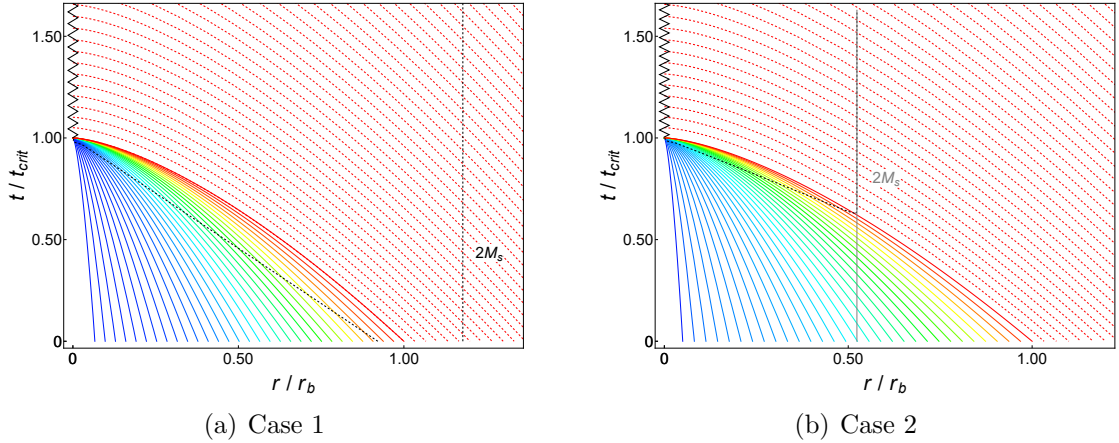


Figure 5.3: Timelike geodesics of dust shells with conserved mass. Color denotes relative values of mass of each dust shell, with the last red solid curve marking the surface of the matter. The dashed red curves are geodesics of dust shells with interior mass M_s . The dashed black curves are the apparent horizons. The singularity that forms at the center is indicated by the zigzag line.

Of course, we must view the trajectories of the radial null worldlines to verify if these do in fact correspond to genuine horizons and singularities which we can see in Fig. 5.4. Due to the free-fall lag of the innermost shells, there is very little initial integrated mass near $r = 0$ and the outgoing null rays there are almost 45° lines. As more and more mass accumulates radially, the gravitational field strengthens and the null geodesics become progressively steeper than 45° . Past the critical time, the entire mass is concentrated at $r = 0$. Thus, the gravitational field strength causes the interior apparent horizon to vanish.

In the first case, the mass has already localized enough to generate the apparent horizon on the initial time slice such that the outgoing null rays will already be trapped. Exterior to the horizon at the Schwarzschild radius of the star, null rays can again escape the gravitational field. Here, it is evident that the trapped region is the region bounded by the exterior horizon and $r = 0$. This verifies that the stationary exterior apparent horizon is in fact the event horizon.

For the second plot, the absence of apparent horizons in the first time slice means that all outgoing null rays will initially be escaping. As the mass becomes localized, we notice that some of the null rays originating from $r < 2M_s$ are able to escape while others become trapped. Thus, the event horizon in this case is traced by some null geodesic originating from $r < 2M_s$ that intersects the boundary of the star exactly when it approaches $r = 2M_s$.

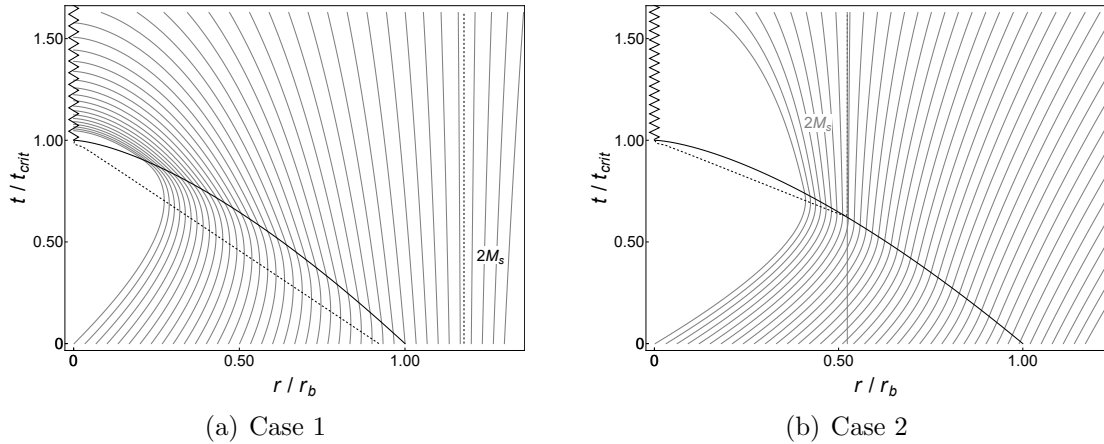


Figure 5.4: Outgoing radial null worldlines propagating outwards from the initial hypersurface. The dashed black curves are the apparent horizons. The black curve is the collapsing matter boundary that turns into a singularity at the center indicated by the zigzag line.

We can also track the outgoing and ingoing null rays emitted from the surface of the star as it collapses, which define the local future light cones or the boundaries between regions accessible to causal observers. This is presented in Fig. 5.5 which shows that the outgoing and ingoing null geodesics indeed converge as they approach the singularity, demonstrating the inevitability of causal observers falling into it. Since the coordinate time is the proper time of the fluid elements, the timelike observers comoving with the surface of the star reach the singularity at some finite time, as opposed to the stationary observers of the Schwarzschild singularity.

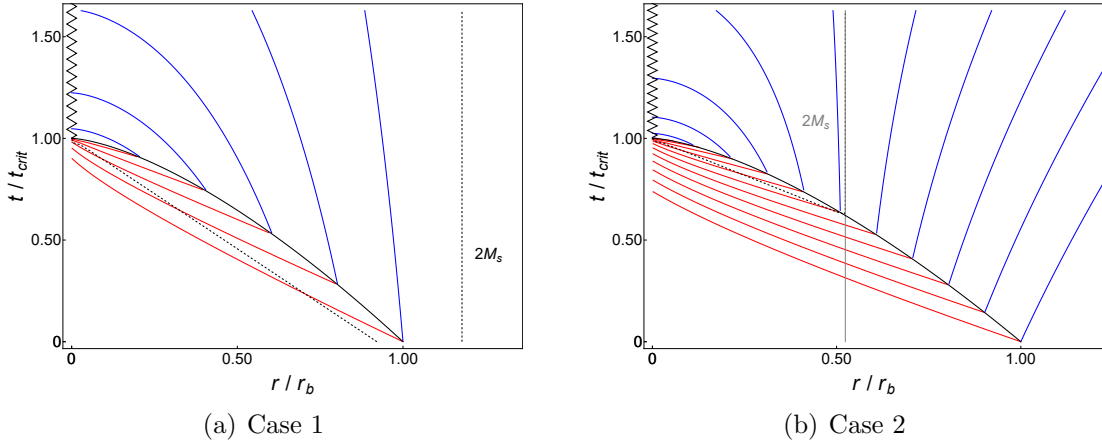


Figure 5.5: Outgoing (blue) and ingoing (red) future radial worldlines of light emitted from the surface of the matter (black) as it collapses.

Revisiting the dependence of the apparent horizon formation on the input parameters, we highlight the fact that each dust shell inside the star has its own conserved mass. As such, we can think of each and every dust shell to all have its own Schwarzschild radius defined by $r_s = 2M_0(r_i) := 2M_i$, where M_i is the conserved mass of the r_i dust shell. Using (5.17), we can derive the time that it takes for the i^{th} dust shell to reach its Schwarzschild radius by equating $r_i(t_{\text{AH}}) = 2M_i$. The apparent horizon formation time of the i^{th} dust shell is given by

$$t_{\text{AH}}(r_i; \rho_c) = \frac{1}{\sqrt{6\pi\rho_c}} - \frac{16}{9}\pi\rho_c r_i^3. \quad (5.20)$$

For a given input of ρ_c , we want to know the minimum value of t_{AH} for $r_i \in (0, r_b]$ so that we know which shell will be the first to collapse to its r_s for a given parameter pair (ρ_c, r_b) . Evidently, this value is minimized when $r_i = r_b^6$ which tells us that the

⁶The extrema is $r_i = 0$, which is the maximum time of AH formation

apparent horizon of OS stars will always originate from its boundary, not at $r = 0$. Moreover, this will allow us to derive a relationship between the input parameters that will define a threshold density given by

$$\rho_c = \frac{3}{8\pi r_b^2}, \quad (5.21)$$

which is the value of density that will give $t_{\text{AH}} = 0$, meaning that the apparent horizon will be at the boundary of the star on the initial hypersurface. Any value of density higher than this will give a negative t_{AH} , meaning there is already an apparent horizon, and any value lower will give some finite time $t_{\text{AH}} > 0$.

Alternatively, we can start by considering that the initial position of a dust shell is related to its conserved mass and initial density by $r_i = \sqrt[3]{3M_i/4\pi\rho_c}$. Similarly, we can equate this to r_s in order to obtain the threshold density that will make $r_i = r_s$. We then obtain $\rho_{\text{crit}} = 3/32\pi M_i^2$ which allows us to define the parametric curve

$$(r_b, \rho_c) = (r_s, \rho_{\text{crit}}) \quad (5.22)$$

with the conserved mass as the parameter. Note that both of these derived conditions are one and the same. Hence, we have a candidate condition that must be the boundary that divides our choices of input parameters into the two cases. We verify this by repeating our numerical simulations across the parameter space and verifying which case they belong to. Due to the implicit nature of the apparent horizon condition, we have to rely on a discrete computational grid to find its dependence on the input parameters. To discriminate between the two cases, we simply have to count the number of roots of (5.14) as it evolves. For example, case 1 occurs when the number of roots starts off as two then eventually becomes one⁷.

We start by choosing a pair of parameters (ρ_c, r_b) and implicitly solve for M across a (t, r) computational grid. At each discrete time t_i , we obtain the vector array $\Delta = r_j - 2M(t_i, r_j)$. The number of roots of (5.14) then at time t_i is the number of times that the sign changes in the components of Δ . We then move to the next time step t_{i+1} and repeat until the last time step. Our computation is neatly summarized in Fig. 5.6⁸. As we can see, the threshold density curve denoted by

⁷Since we are doing discrete calculations, we do not expect to be able to capture the exact time when the apparent horizon first appears.

⁸This is a discrete 20×20 grid of pairs (ρ_c, r_b) with discrete values ranging from $\rho_c \in (0.1, 1)M_\odot^{-2}$ and $r_b \in (0.1, 1.25)M_\odot$.

the white dashed curve exactly lines up with the separation of cases in the discrete parameter pair grid, verifying our claim.

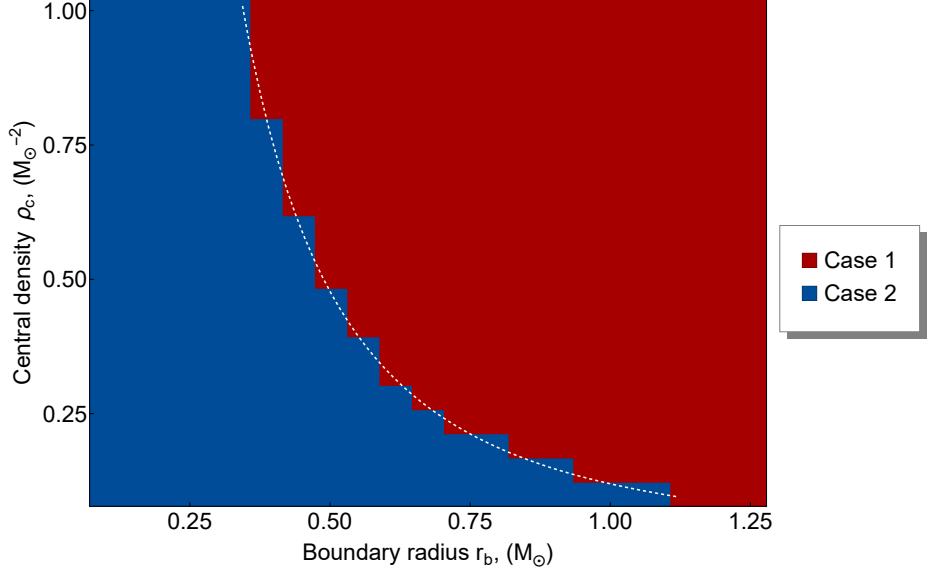


Figure 5.6: Qualitative classification of the apparent horizon formation of the Oppenheimer-Snyder spacetime depending on its parameter space. The white dashed curve is Eq. (5.21).

In order to setup the analysis later, we illustrated in Fig. 5.7 the contour curves of t_{AH} for values of r_i and r_b , given a constant value of $\rho_c = 0.25M_\odot^{-2}$. Here, we excluded the region $r_i > r_b$ as this region is outside the star and hence is not matter-filled. Since (5.21) does not depend on r_b notice that t_{AH} is constant along vertical lines. By choosing some constant slice of r_b , say $r_b = 0.25M_\odot$, this contour plot tells us the time in which the dust shell originating from r_i will collapse into its Schwarzschild radius for this dust collapse model with input parameters $(\rho_c, r_b) = (0.25M_\odot^{-2}, 0.25M_\odot)$. From the plot, the boundary shell always has the lowest value of t_{AH} and we again conclude that an apparent horizon always forms when the star surface approaches the Schwarzschild radius of the star. We note that the dashed line that separates r_b values which admit positive and negative t_{AH} values is derived from (5.21) for the input of $\rho_c = 0.25M_\odot^{-2}$.

As a brief aside, we wish to compute the scale factor $a(t)$ of the equivalent FLRW spacetime of the collapsing star model. While not essential to the main analysis, this calculation provides a nice verification that we are in fact dealing with an FLRW spacetime. It is essential to point out that along each geodesic, $\rho_c(t)r_i(t)^3 = \text{constant}$. By equating this to their initial values, we can find the explicit form of $\rho_c(t)$ which is

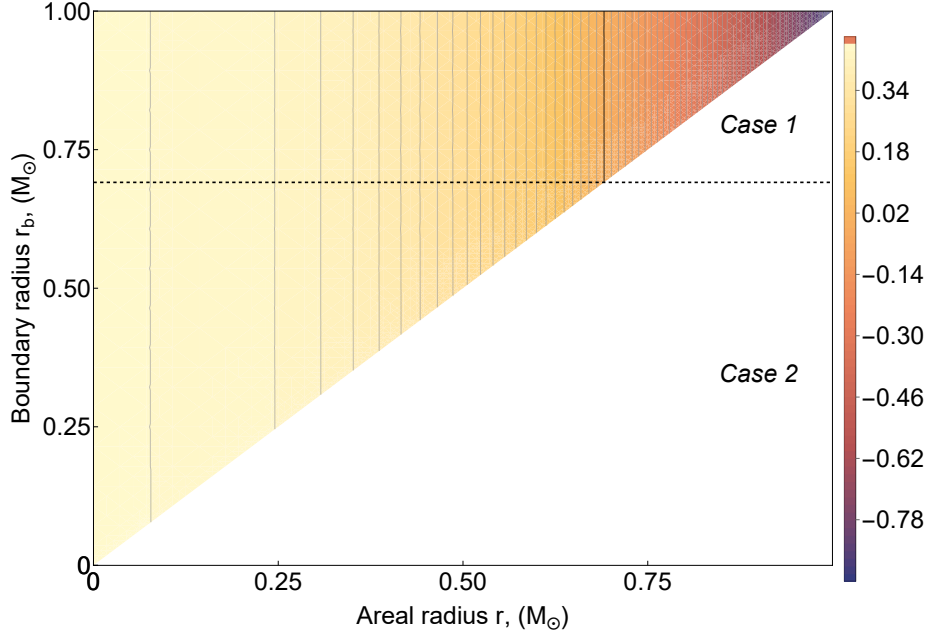


Figure 5.7: Contour plot of $t_{\text{AH}}(r)$ for $\rho_c = 0.25 M_\odot^{-2}$. The black contour line is $t_{\text{AH}} = 0$. The dashed line demarcates the boundary of r_b that separate each case in Fig. 5.6. The region $r > r_b$ is excised.

given by

$$\rho_c(t) = \frac{\rho_c(0)}{(1 - t/t_{\text{crit}})^2} \sim t^{-2}. \quad (5.23)$$

This will allow us to compute the scale factor using Friedman equation, which gives us

$$a(t) = \left(1 + \frac{t}{t_{\text{crit}}}\right)^{-2/3} \sim t^{-2/3} \quad (5.24)$$

where we have set $a(0) := 1$. These solutions exactly correspond to (3.10) with $\omega = 0$, which is a collapsing, dust, and flat Friedman universe, which gives us a good sanity check that this exactly an FLRW spacetime.

While this dynamical collapse model is interesting, it is, of course, an idealized scenario involving a constant density star. In reality, stars develop dense cores with radially varying density profiles. Our model is most apt to describe a large cloud of homogeneously distributed, non-interacting cold dark matter. The most “realistic” scenario captured by this model would be an extremely low density dust cloud spread over an extremely large region. The collapse scenario would then happen over an extremely longer period of time. Even then, the assumption of homogeneity is still

highly idealized, as gravitational collapse implies that matter becomes increasingly localized. Intuitively, the self-gravity of individual dust particles would lead to the formation of a higher density central region long before a full-scale collapse occurs. Due to these dense central cores, we should expect the singularity to originate at the center of the dust cloud, rather than forming at some finite radius.

Now that we have established the framework for analyzing collapse in this simplified case, we are equipped to extend our analysis to more realistic stellar configurations with non-uniform density profiles.

5.2 Inhomogeneous dust collapse

To gain intuition and keep the analysis tractable, we begin by assuming a simple linear form for the initial density. This choice serves as a baseline model that captures the essential features while remaining analytically manageable. We parametrize the initial energy density profile in the form:

$$\rho_0(r) = \begin{cases} \rho_c \left(1 - \frac{r}{r_b}\right) & \text{if } r \leq r_b, \text{ and} \\ 0 & \text{if } r > r_b \end{cases} \quad (5.25)$$

Now, we have a linearly decreasing $\rho_0(r)$ with central density ρ_c that again cuts off at the stellar boundary. Since ρ_0 is now a genuine function of the coordinate radius, the interior region is now described by the LTB geometry. The initial mass would then be

$$M_0(r) = 4\pi\rho_c \times \begin{cases} \left(\frac{r^3}{3} - \frac{r^4}{4r_b}\right) & \text{if } r \leq r_b, \text{ and} \\ \frac{r_b^3}{12} & \text{if } r > r_b \end{cases} \quad (5.26)$$

and is visualized in Fig. 5.8. Again, evolving it according to the field equation yields the same qualitative behavior; with the mass function indicating a collapse to $r = 0$ and which scales appropriately with the input parameters. However, unlike the OS collapse, the dust shells here collapse more gradually, with their infall spread out over time. The innermost shell still appears to reach $r = 0$ at $t = t_{\text{crit}}$ and the entire collapse process concludes at $t_{\text{coll}} = 2t_{\text{crit}}$. This suggests that an inhomogeneous dust distribution with the same central density as a homogeneous one results in twice the overall collapse time.

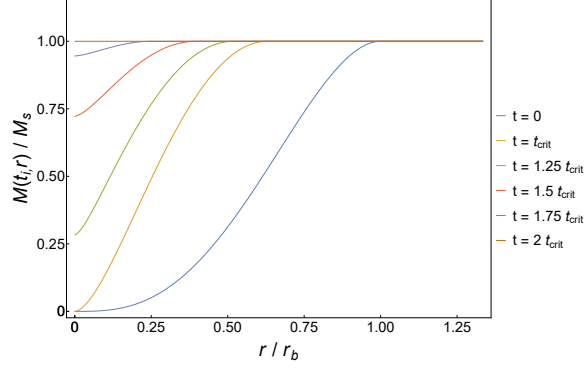


Figure 5.8: Mass evolution of the linearly inhomogeneous matter

More interestingly however, there are now three scenarios for the formation of the apparent horizons;

1. Two initial AHs; the outer horizon is at the Schwarzschild radius and the inner horizon gradually shrinks until it approaches $r = 0$ at $t = t_{\text{crit}}$.
2. No initial AHs, then one appears somewhere inside the star. The horizon then splits, with the outer horizon expanding outwards into $r = 2M_s$ and the inner horizon disappears at $t = t_{\text{crit}}$.
3. No initial AHs until $t = t_{\text{crit}}$, then one forms from $r = 0$ and expands outwards until it reaches $r = 2M_s$.

The first two are almost exactly the same as the OS collapse case, with the caveat in case 2 that the dust shell responsible for the initial formation of the apparent horizon is *not always* the surface of the star. A more interesting qualitative result is the emergence of case 3, which is exactly what we would expect for a realistic collapse. Just as before, we reserve the discussion for later.

The evolution of the mass function along with the respective apparent horizons is illustrated in Fig. 5.9, again for each of the cases. As before, case 1 describes the scenario where the star is already inside its Schwarzschild radius and is almost exactly identical with the OS case⁹. It can be more readily seen in the second case that the apparent horizon has formed somewhere within the star. Lastly, we clearly see the slow evolution of the horizon in case 3 and its emergence from the center before settling in on the Schwarzschild radius.

⁹Moving forward, we exclude the analysis of case 1 to avoid redundancy. However, we will still include the corresponding plots for the sake of completeness

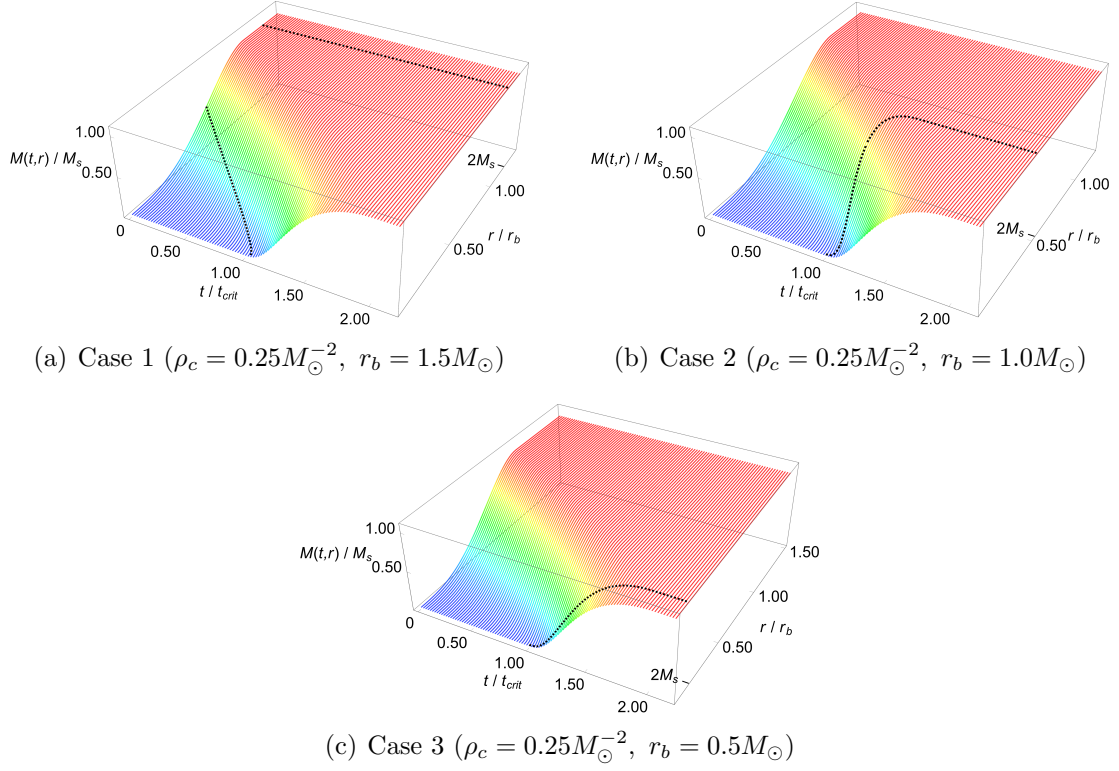


Figure 5.9: Time evolution of the linearly inhomogeneous matter mass function for three different input parameters. Plot elements are as described in Fig. 5.2

From the figures above, one might be inclined to believe that the total collapse time in this scenario is $t_{\text{coll}} = 2t_{\text{crit}}$. We compute for the worldlines of the i^{th} dust as given below

$$r_i(t) = \begin{cases} r_i \left(1 - \frac{t}{t_{\text{crit}}} \sqrt{1 - \frac{3r_i}{4r_b}} \right)^{2/3} & \text{if } r_i \leq r_b, \text{ and} \\ r_i \left(1 - \frac{t}{2t_{\text{crit}}} \frac{r_b^{3/2}}{r_i^{3/2}} \right)^{2/3} & \text{if } r_i > r_b, \end{cases} \quad (5.27)$$

which verifies that the boundary of the star does in fact approach $r = 0$ at $t_{\text{coll}} = 2t_{\text{crit}}$. The geodesic deviation vector of the dust shells are written as

$$\xi^a = \frac{1 - \frac{\lambda}{t_{\text{crit}}} \frac{1 - r_i/r_b}{\sqrt{1 - 3r_i/4r_b}}}{\left(1 - \frac{\lambda}{t_{\text{crit}}} \sqrt{1 - \frac{3r_i}{4r_b}} \right)^{1/3}} (\partial_r)^a \quad (5.28)$$

Again, the geodesics only vary along the radial direction and are indicative of spherical collapse. The geodesic deviation equation then reads:

$$R_{bcd}^a n^b n^c \xi^d = - \frac{6\mathcal{R}_1 \sqrt{\mathcal{R}_2} - 4\mathcal{R}_2 \sqrt{1 - \frac{3r_b r_i}{4}} - \frac{2t}{t_{\text{crit}}} \mathcal{R}_1 \mathcal{R}_2}{9r_b^4 t_{\text{crit}}^4 \sqrt{1 - \frac{3r_b r_i}{4}} \left(1 - \frac{t}{t_{\text{crit}}} \sqrt{\mathcal{R}_2}\right)^{7/3}} (\partial_r)^a. \quad (5.29)$$

where $\mathcal{R}_1 = 1 - \frac{r_i}{r_b}$ and $\mathcal{R}_2 = 1 - \frac{3r_i}{4r_b}$. We evaluate this relation at $r_i = 0$ to find the tidal strength caused by the spacetime curvature at the stellar center, which is

$$R_{bcd}^a n^b n^c \xi^d \Big|_{r_i \rightarrow 0} = -\frac{2}{9} \frac{1}{t_{\text{crit}}^4} \left(1 - \frac{t}{t_{\text{crit}}}\right)^{-4/3} (\partial_r)^a. \quad (5.30)$$

Again, we observe that the spacetime curvature diverges past the critical time which is again indicative of a genuine physical singularity at $r = 0$. Another way to observe the presence of the geometric singularity is to analyze the timelike geodesics is by inverting the expression:

$$t(r; r_i) = t_{\text{crit}} \times \begin{cases} \sqrt{\frac{4r_b}{4r_b - 3r_i}} \frac{1}{r_i^{3/2}} \left(r_i^{3/2} - r^{3/2}\right) & \text{if } r_i \leq r_b, \text{ and} \\ \frac{2}{r_b^{3/2}} \left(r_i^{3/2} - r^{3/2}\right) & \text{if } r_i > r_b. \end{cases} \quad (5.31)$$

By evaluating this at $r = 0$, as shown below, we find the proper time measured by each dust shell as it collapses from its initial position r_i to $r = 0$. Now we more clearly see that the innermost shell ($r_i = 0$) will collapse at t_{crit} , followed by the rest of the contents of the star, until lastly the boundary ($r_i = r_b$) collapses at $t_{\text{coll}} = 2t_{\text{crit}}$.

$$t(0; r_i) = t_{\text{crit}} \sqrt{\frac{4r_b}{4r_b - 3r_i}} \quad (5.32)$$

The spacetime diagram of the linear inhomogeneous model is depicted in Fig. 5.10. Unlike the sudden collapse of the entire star at the critical time in the OS case, we observe here the gradual collapse of each dust shell into the center from $t = t_{\text{crit}}$ to $2t_{\text{crit}}$. Note that the singularity appears within the star itself at the critical time. We again observe the same free-fall lag for the central dust shells. As before, the parameters first case initializes a black hole on the initial time slice while the other two choices contain stars with its Schwarzschild radius either being in some intermediate radius (case 2) or very near the center (case 3).

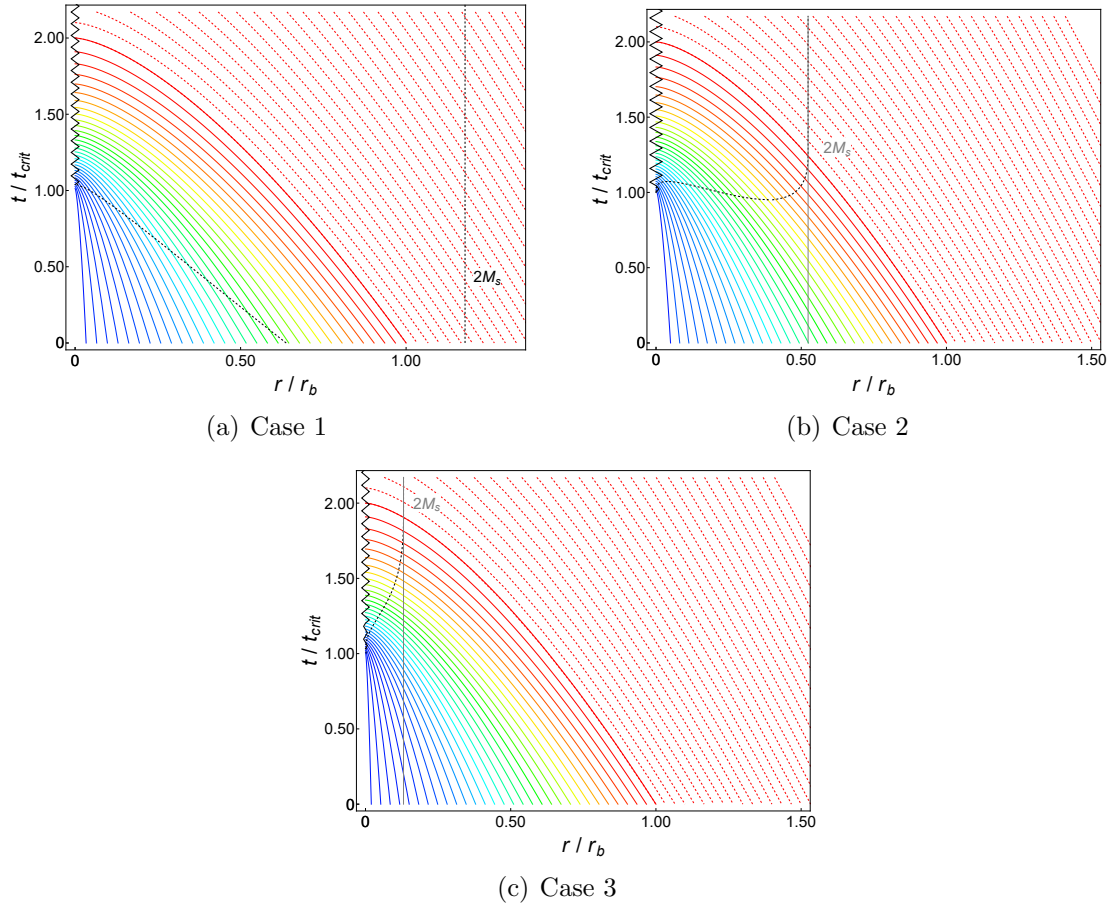


Figure 5.10: Timelike geodesics for the linearly inhomogeneous matter. Plot elements are as described in Fig. 5.3.

Tracking the outgoing null worldlines originating from the initial hypersurface as illustrated in Fig. 5.11 verify the absence of a horizon for cases 2 and 3. The second case behaves similarly to its OS counterpart, with the genuine event horizon being traced by some outgoing null geodesic originating from $r < 2M_s$ that intersects that boundary of the star exactly at $r = 2M_s$. In case 3, we observe that the gravitational field strength of the collapsing matter is much weaker compared to the first two cases as evident from the innermost outgoing null ray being able to completely escape to infinity. Hence, we expect that its event horizon will appear at $r = 0$ at some later time $0 < t < t_{\text{crit}}$. The relatively weak gravitational field in this case follows from the smaller total mass of the star as a consequence of the choice of parameters.

Lastly, tracking the emitted light from the surface of the star as it collapses in Fig. 5.12 do verify the future existence of its horizons. Reiterating that the red and blue curves define the local future light cones, the inevitability of observers approach-

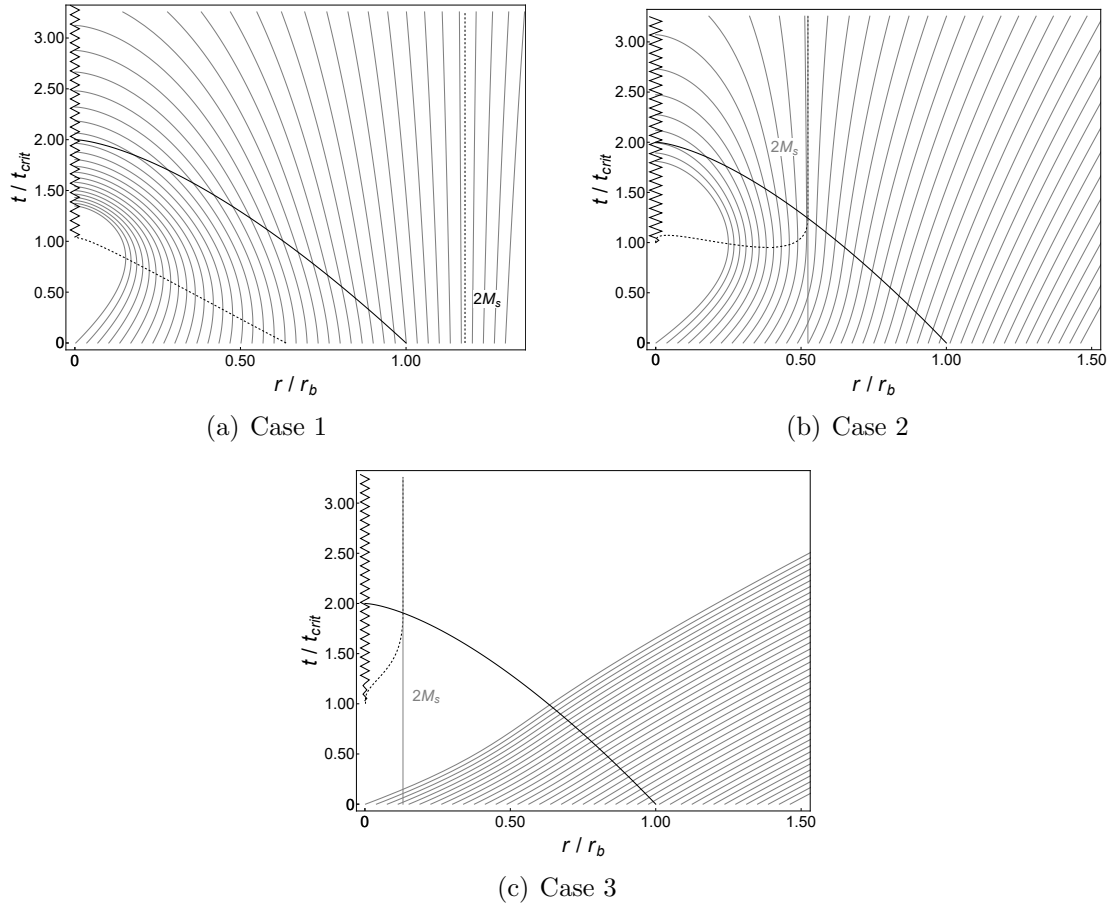


Figure 5.11: Outgoing radial null worldlines of the linearly inhomogeneous matter propagating outwards from the initial hypersurface. Plot elements are as described in Fig. 5.4

ing the singularity upon entering the horizons becomes apparent. It is also evident that both outgoing and ingoing null rays converge at the singularity.

Probing the parameter space, we create another 20×20 discrete grid of parameter pairs (ρ_c, r_b) and distinguish between the three cases as presented in Fig. 5.13. This time, the number of roots of (5.14) on the initial time slice is zero for both cases 2 and 3. Thus, the way we distinguish between cases is by storing the progression of the number of apparent horizons for increasing time slices. For example, the number of roots for case 2 will progress as $0 \rightarrow 2 \rightarrow 1$, while case 3 will progress as $0 \rightarrow 1$. We then color the grid according to the legend, noting that the parameter space is neatly divided into three distinct regions, each corresponding to one of the three cases of apparent horizon formation behavior. We now attempt to understand why the parameter space is split as such.

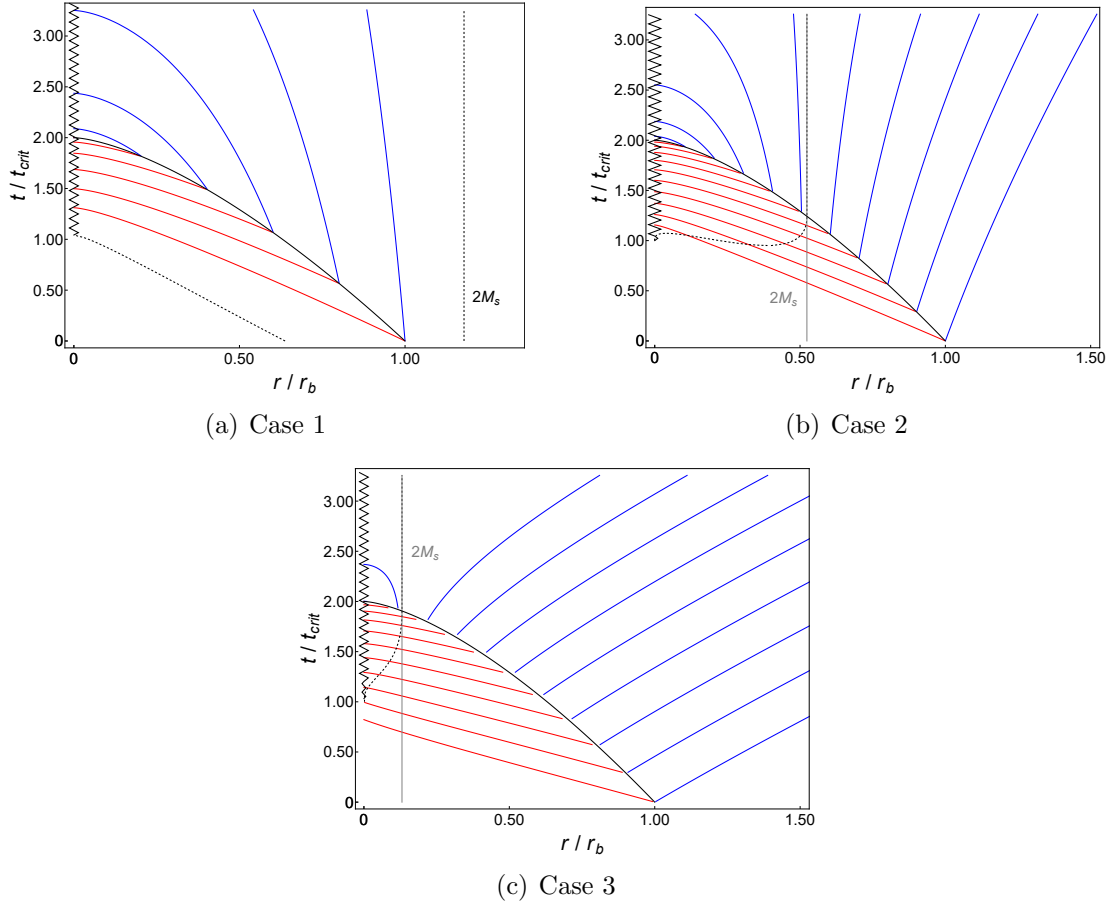


Figure 5.12: Outgoing (blue) and ingoing (red) future radial worldlines of light emitted from the surface of the linearly inhomogeneous matter (black) as it collapses.

We derive the threshold density in which the inhomogeneous matter is compact enough to already be inside its Schwarzschild radius. We find this by equating the initial boundary radius with the Schwarzschild radius. In this case, it is given by $\rho_{\text{crit}} = 3/8\pi M_s^2$ which allows us to again obtain the parametric curve

$$(r_b, \rho_c) = (r_s, \rho_{\text{crit}}) \rightarrow \rho_c = \frac{3}{2\pi r_b^2}, \quad (5.33)$$

An important point to note is that the threshold initial central density required for an initial BH configuration in an inhomogeneous distribution is higher than in the homogeneous case. This is expected, since the energy is more spatially spread out, allowing the system to sustain a higher central density before fully localizing into a singularity. Overlaying this curve onto Fig. 5.13 verifies that this does indeed separate the parameter space into choices that lead to cases 1 and 2. In order to split the remaining parameter choices between cases 2 and 3, we progress by consulting

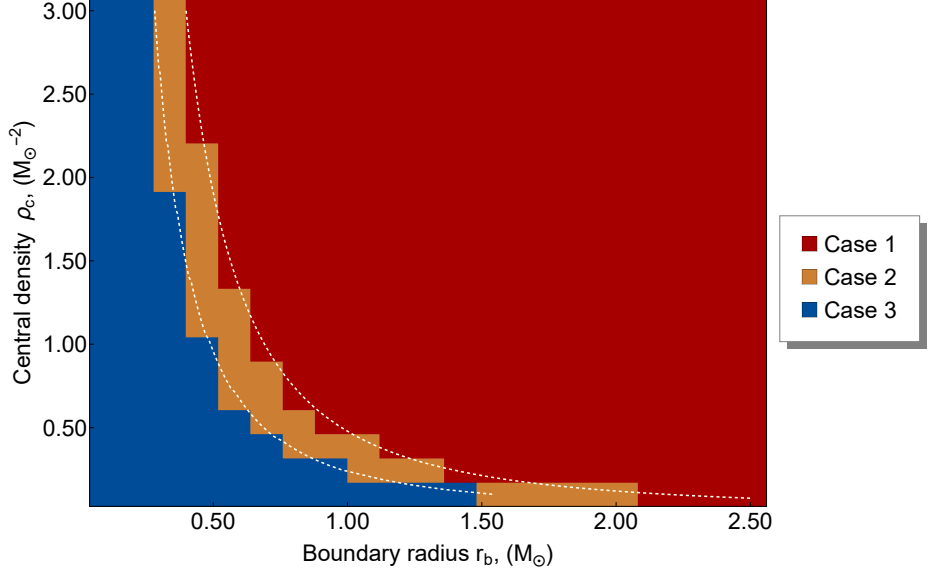


Figure 5.13: Qualitative classification of the apparent horizon formation of the linearly inhomogeneous matter spacetime depending on its parameter space.

the t_{AH} for this configuration.

The apparent horizon formation time for the i^{th} dust shell in the linear inhomogeneous model is

$$t_{\text{AH}}(r_i; \rho_c, r_b) = \frac{1}{\sqrt{6\pi\rho_c(1 - 3r_i/4r_b)}} - \frac{16\pi\rho_c r_i^3}{9} \left(1 - \frac{3r_i}{4r_b}\right). \quad (5.34)$$

Recall that this expression is obtained by substituting t_{AH} into (5.27) and solving for it from $r_i(t_{\text{AH}}) = 2M_i$. This expression becomes complex-valued for $r > 4r_b/3$, which is not a concern since this is only valid for $r \leq r_b$. For input parameters (ρ_c, r_b) , the minima of t_{AH} in the region $r_i \in (0, r_b]$ will tell us which dust shell will first form the apparent horizon, whether outside the star (case 1), inside the star (case 2), or the innermost shell (case 3). Note that the form of the first derivative of (5.34) with respect to r is

$$\frac{dt_{\text{AH}}}{dr} \sim r^3 + r^2 + \frac{\sqrt{1-r}}{r^2 + r} \quad (5.35)$$

and hence its extrema are not analytically solvable and we rely on numerical computations. To find the minimum values of (5.34), we substitute $\rho_c = 0.25M_{\odot}^{-2}$ to reduce its dimensionality and plot its contours which is illustrated in Fig. 5.14. Noting that the black contour line corresponds to $t_{\text{AH}} = 0$, we verify that case 1 can be separated from cases 2 and 3 by using (5.33) and substituting $\rho_c = 0.25 M_{\odot}^{-2}$. By

inspecting the contours, we observe that for small r_b values, the function $t_{\text{AH}}(r)$ is minimized at $r = 0$, indicating that the innermost dust shell collapses first into its Schwarzschild radius. This implies that the apparent horizon originates at the center and propagates outward toward $r = 2M_s$, as shown in Fig. 5.14 by the increasing values of t_{AH} with radius.

For intermediate values of r_b , however, we find that t_{AH} is minimized at some finite radius $0 < r < r_b$, evidenced by the parabolic shape of the contours. This suggests that in this region, the apparent horizon first forms at a finite radius within the star and subsequently splits into interior and exterior segments as time progresses, which is consistent with the parabolic structure of the t_{AH} contours.

The key insight for distinguishing the two regions in the parameter space (Fig. 5.13) is to determine the smallest value of r_b for which t_{AH} is no longer minimized at $r = 0$. To do this, we scan over $N = 100$ discrete values of ρ_c and numerically identify the corresponding threshold values of r_b . This procedure yields the dashed white curve in Fig. 5.13, which indeed separates the parameter choices leading to cases 2 and 3.

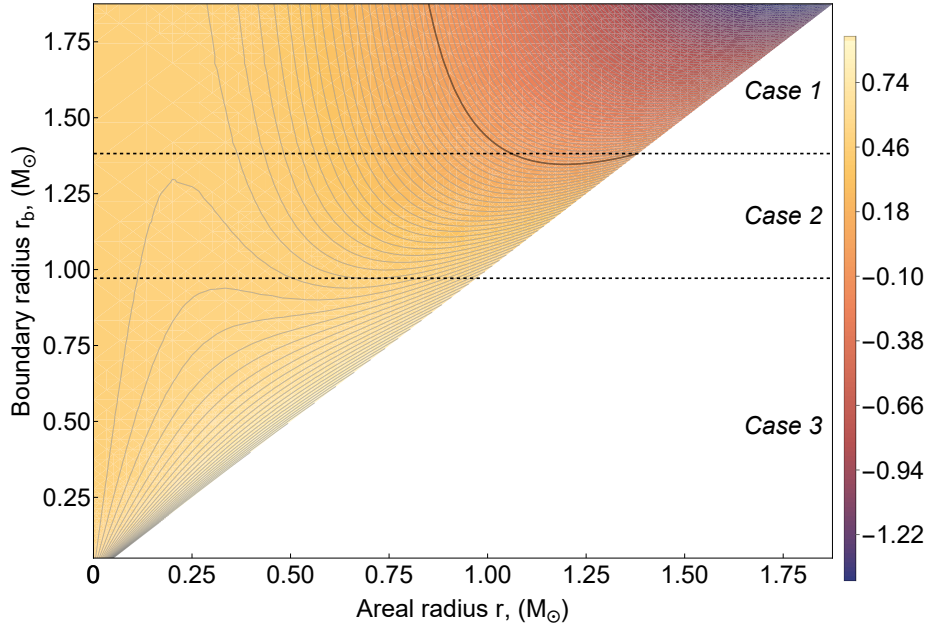


Figure 5.14: Contour plot of $t_{\text{AH}}(r; r_b)$ for $\rho_c = 0.25 M_{\odot}^{-2}$. The black contour line is $t_{\text{AH}} = 0$. The black dashed lines demarcate the boundaries of r_b that separate each case in Fig. 5.13. The region $r > r_b$ is excised.

Now, a more realistic configuration of stellar energy density is a Gaussian centered at $r = 0$. Its gentle descent to lower values reflect a more “physical” density profile. One could compare this profile to the solutions of the Lane-Emden equation, which

correspond to stable, self-gravitating, polytropic fluids that are widely used as great approximations of stable stellar structure [11]. We can think of this model as a star at the end of its lifetime once it has used up all of its nuclear fuel and can no longer sustain internal pressure through fusion reactions. At this stage, the star is supported only by residual pressure sources (such as electron or neutron degeneracy pressure) which we shall assume here to be negligible. The Gaussian energy density profile thus serves as an approximate model for such compact stellar remnants, where the mass is still concentrated near the center, but the pressure is no longer sufficient to counteract gravity throughout the entire volume. We parametrize the density as follows:

$$\rho_0(r) = \rho_c \exp \left[-\frac{1}{2} \left(\frac{r}{\sigma} \right)^2 \right]. \quad (5.36)$$

Here, σ is the characteristic width of the density profile, or how spread apart it is initially. Since Gaussian functions do not vanish at a finite radius we define an effective boundary $r_b = 3\sigma^{10}$ in which we consider values greater than r_b to be the vacuum region. The effective Schwarzschild mass observed by distant observers is $M_s = \lim_{r \rightarrow \infty} M_0(r) = (2\pi)^{3/2} \rho_c \sigma^3$. Using the matter relation again, the initial mass function would be

$$M_0(r) = 4\pi\rho_c \left[\sqrt{\frac{\pi}{2}} \sigma^3 \operatorname{erf} \left(\frac{r}{\sqrt{2}\sigma} \right) - r\sigma^2 \exp \left(-\frac{r^2}{2\sigma^2} \right) \right], \quad (5.37)$$

where $\operatorname{erf}(x)$ is the error function, which arises naturally from the integral of a Gaussian function. We again evolve it according to the field equation solution which is shown in Fig. 5.15. Just as the other inhomogeneous case, we see that the collapse happens more gradually than the homogeneous OS case, where it finalizes at around $t = 3 - 4t_{\text{crit}}$.

Interestingly, we find that the collapse of this star with an initial Gaussian energy density is qualitatively similar to the linearly inhomogeneous case. This becomes evident upon consulting the apparent horizon formation behavior as seen in the mass function surface plots in Fig. 5.17. We observe the exact same qualitative results, where we see the same three distinct apparent horizon formation behaviors; one where the star is already inside its Schwarzschild radius and has two apparent horizons on the initial time slice, one where the apparent horizon forms at some finite radius and

¹⁰This choice is completely arbitrary, and this choice was done so that we do not need to track large values of t and r in our computational grid.

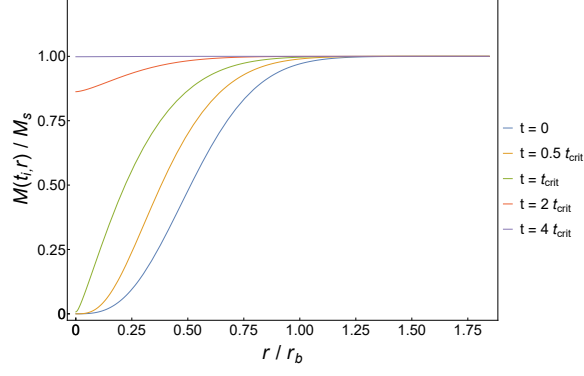


Figure 5.15: Mass evolution of the Gaussian inhomogeneous matter

splits into the vanishing interior horizon and exterior horizon at the Schwarzschild radius, and lastly one where the apparent horizon forms from the center of the star at the critical time and extends outwards to its Schwarzschild radius. A common feature across all remaining plots is that adopting a more physically motivated initial mass function results in smoother gradients and more regular horizon structures.

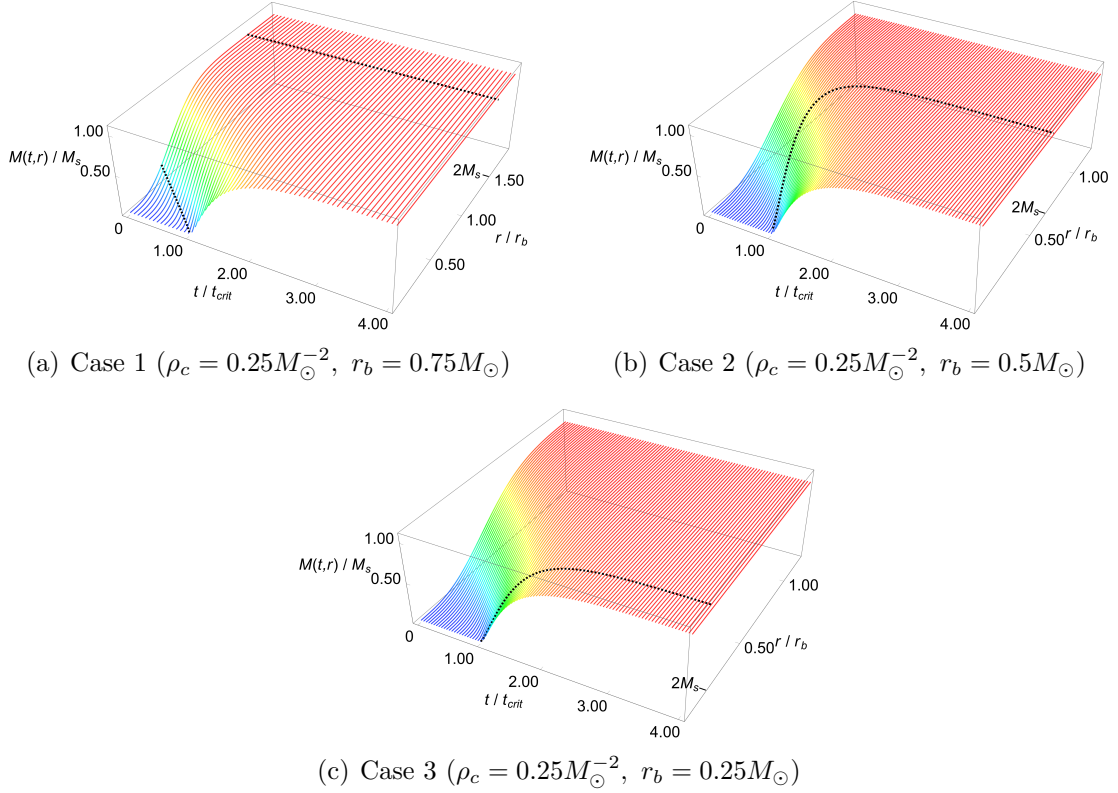


Figure 5.16: Time evolution of the Gaussian inhomogeneous matter mass function for three different input parameters. Plot elements are as described in Fig. 5.2

We want to know the total collapse time of the i^{th} dust shell situated at some $r_i = N\sigma$. The timelike geodesics here is given by:

$$r(t; r_i) = \left(r_i^{3/2} - 3t\mathcal{T}(r_i) \right)^{2/3}, \quad (5.38)$$

where $\mathcal{T}(r_i) = \sqrt{\pi\rho_c\sigma^2 \left(\sqrt{2\pi}\sigma \operatorname{erf}\left(\frac{r_i}{\sqrt{2}\sigma}\right) - 2r_i e^{-r_i^2/2\sigma^2} \right)}$. Notice that this function is actually the critical time in disguise, as taking the limit shows that $\lim_{r_i \rightarrow \infty} \mathcal{T}(r_i) = \sqrt{(2\pi)^{1/2}\pi\rho_c\sigma^3} \sim t_{\text{crit}}^{-1}$. The collapse time of the i^{th} dust shell is

$$t_{i,\text{coll}}(N) = \frac{N^{3/2}}{3\sqrt{\pi\rho_c \left(\sqrt{2\pi} \operatorname{erf}\left(\frac{N}{\sqrt{2}}\right) - 2N e^{-N^2/2} \right)}}, \quad (5.39)$$

which we have obtained by taking the time where $r(t_{i,\text{coll}}; r_i) = 0$. Since we arbitrary chose our star boundary such that $r_b = 3\sigma$, the total collapse time of our matter configuration is

$$\begin{aligned} t_{\text{coll}} &= \sqrt{\frac{3}{\pi\rho_c \left(\sqrt{2\pi} \operatorname{erf}\left(\frac{3}{\sqrt{2}}\right) - \frac{6}{e^{9/2}} \right)}} \\ &\approx \sqrt{\frac{3}{\sqrt{2\pi^3\rho_c}}} \approx 2.68 t_{\text{crit}}, \end{aligned} \quad (5.40)$$

noting that $\operatorname{erf}(3/\sqrt{2}) \approx 1$ and $e^{-9/2} \approx 0$. The deviation vector now reads

$$\xi^a = \left(\sqrt{r_i} - \frac{2e^{-r_i^2/2\sigma^2} r_i^2 \lambda \rho_c}{\mathcal{T}(r_i)} \right) / \left(r_i^{3/2} - 3\lambda \mathcal{T}(r_i) \right) (\partial_r)^a, \quad (5.41)$$

noting that $\lambda \mathcal{T}(r_i) \sim t/t_{\text{crit}}$. The geodesic deviation equation then reads:

$$R_{bcd}^a n^b n^c \xi^d = \frac{4\pi\rho_c \sqrt{r_i} e^{-r_i^2/2\sigma^2} \mathcal{R}_3}{\left(r_i^{3/2} - 3t\mathcal{T}(r_i) \right)^{7/3}} (\partial_r)^a. \quad (5.42)$$

where $\mathcal{R}_3 = \left(-r_i^3 - 2r_i\sigma^2 + e^{r_i^2/2\sigma^2} \sqrt{2\pi}\sigma^3 \operatorname{erf}\left(\frac{r_i}{\sqrt{2}\sigma}\right) + r_i^{3/2} t \mathcal{T}(r_i) \right)$. Evaluating this expression at $r_i = 0$ leads to indeterminate forms (0/0) that cannot be made to vanish with L'Hopital's rule. Taking its Taylor series expansion around $r_i = 0$, we find that

$$R_{bcd}^a n^b n^c \xi^d \Big|_{r_i \rightarrow 0} = \left(-\frac{4\pi\rho_c}{3 \left(1 - \frac{t}{t_{\text{crit}}} \right)^{4/3}} + \mathcal{O}(r_i) \right) (\partial_r)^a. \quad (5.43)$$

Amazingly, we again recover the fact that the spacetime curvature diverges past the critical time which, again, is indicative of a genuine physical singularity at $r = 0$. We present the spacetime diagrams for the Gaussian inhomogeneous model in Fig. 5.17. The same qualitative features we have already observed from the previous inhomogeneous case are still present here; the free-fall lag near $r = 0$ and the more gradual collapse of the matter until $t_{\text{coll}} \approx 2.68t_{\text{crit}}$.

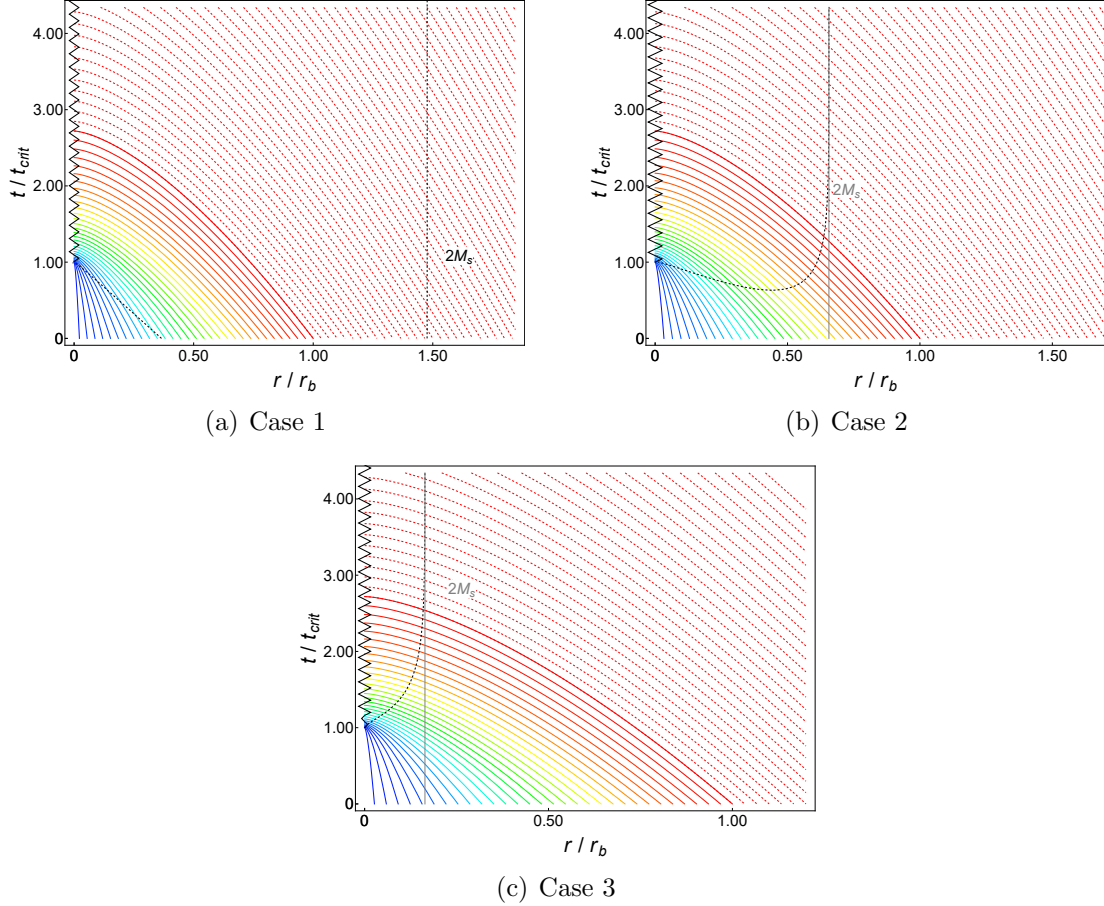


Figure 5.17: Timelike geodesics for the Gaussian inhomogeneous matter. Plot elements are as described in Fig. 5.3.

We again track the outgoing null worldlines from the initial time slice, as shown in Fig. 5.18. In the first case, the stationary exterior apparent horizon coincides with the genuine event horizon of the spacetime, since all outgoing null rays originating within that radius cannot escape to infinity. In the second case, we observe that the event horizon is traced by a null geodesic originating from a point with $r < 2M_s$, which intersects the star's surface precisely as it collapses to its Schwarzschild radius. Lastly, in the third case, the relatively smaller total mass of the star, and consequently the

weaker gravitational field, allows the innermost outgoing null ray to escape to infinity.

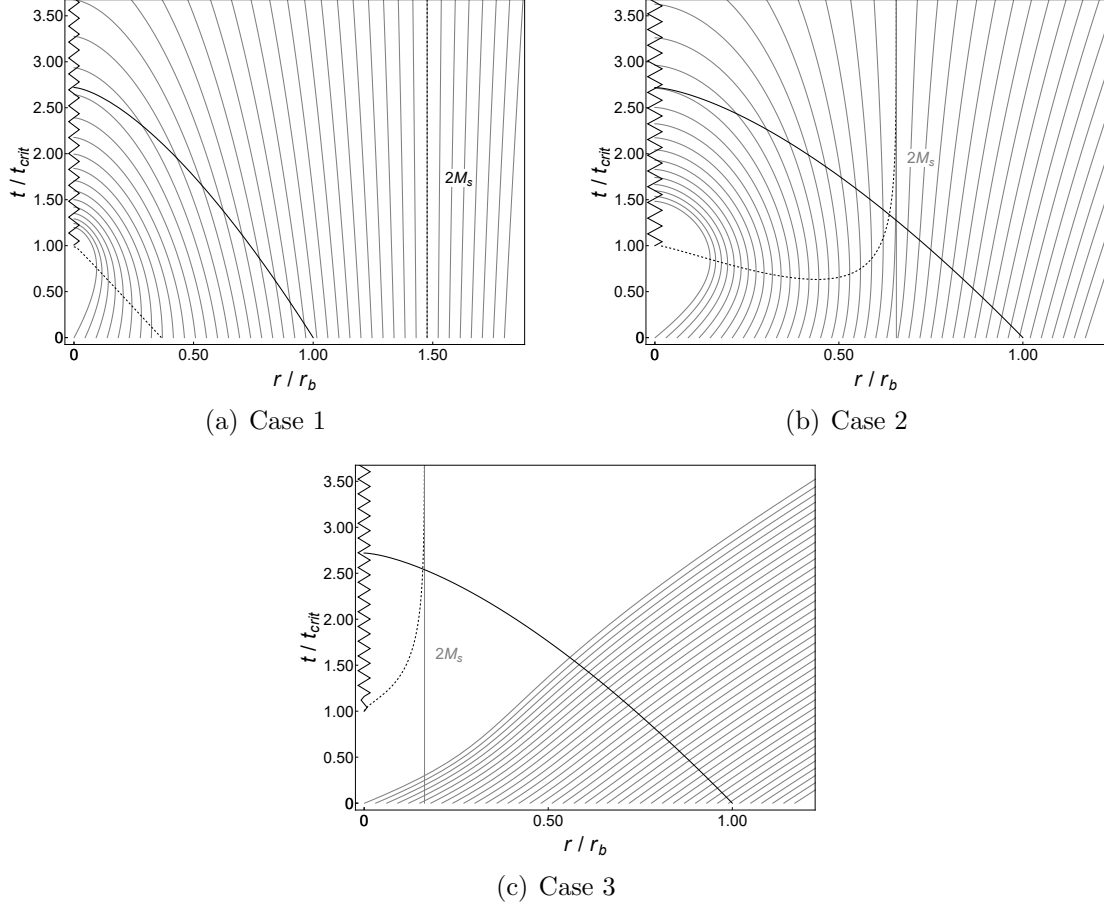


Figure 5.18: Outgoing radial null worldlines of the Gaussian inhomogeneous matter propagating outwards from the initial hypersurface. Plot elements are as described in Fig. 5.4

For the sake of completeness, we also track the light rays emitted from the collapsing star boundary to visualize the local future light cones of causal observers comoving with the stellar surface. As before, the convergence of the outgoing and ingoing null rays becomes evident as the star boundary approaches the singularity.

We once again explore the parameter space, following the same procedure as in the linear case, as illustrated in Fig. 5.20. The resulting qualitative picture remains the same: the parameter space is divided into three distinct regions, each corresponding to a different apparent horizon formation behavior. The threshold density condition that separates case 1 from cases 2 and 3 is again derived, which we have found to be

$$\rho_c = \frac{3}{2(2\pi)^{3/2}\sigma^2}. \quad (5.44)$$

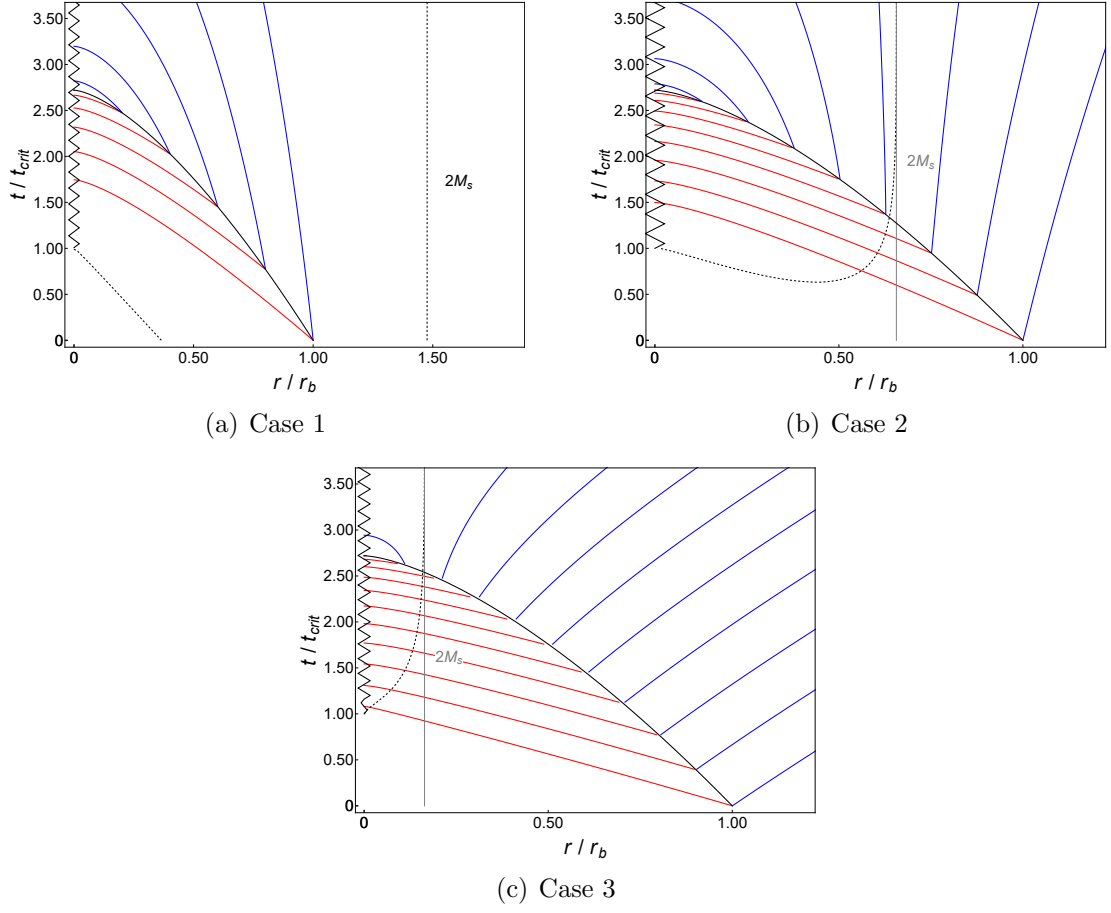


Figure 5.19: Outgoing (blue) and ingoing (red) future radial worldlines of light emitted from the surface of the Gaussian inhomogeneous matter (black) as it collapses.

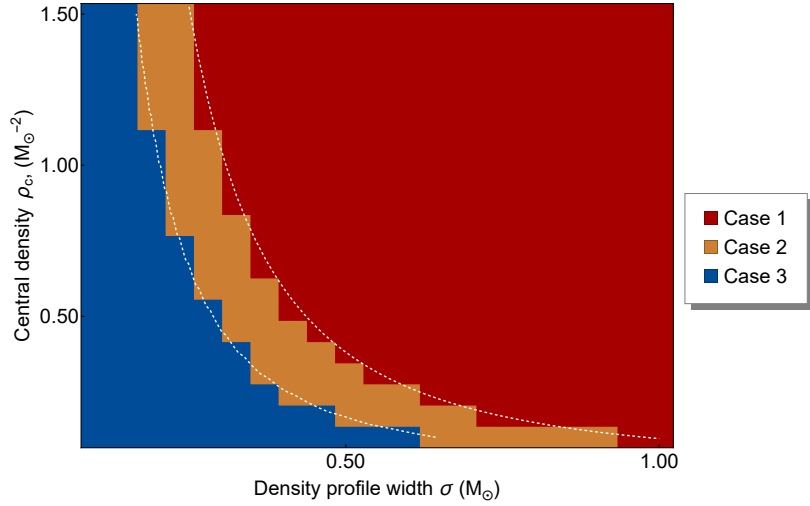


Figure 5.20: Qualitative classification of the apparent horizon formation of the Gaussian inhomogeneous matter spacetime depending on its parameter space.

As before, we compute t_{AH} (shown below) to distinguish between cases 2 and 3. This quantity depends on r_i , as well as the parameters ρ_c and σ . We fix $\rho_c = 0.25 M_{\odot}^{-2}$ and plot the resulting contours in Fig. 5.21, which qualitatively resemble those of the linear case. Parameter choices corresponding to case 3 occur when t_{AH} is minimized at $r = 0$, while case 2 occurs when the minimum is at some finite $0 < r < r_b$, indicated by the parabolic-shaped contours.

$$t_{\text{AH}}(r_i; \rho_c, \sigma) = \frac{8}{3} \pi \rho_c \sigma^2 \left(2r_i e^{-r_i^2/2\sigma^2} - \sqrt{2\pi} \sigma \text{erf} \left(\frac{r_i}{\sqrt{2}\sigma} \right) \right) + \frac{r_i^{3/2}}{3\mathcal{T}(r_i)} \quad (5.45)$$

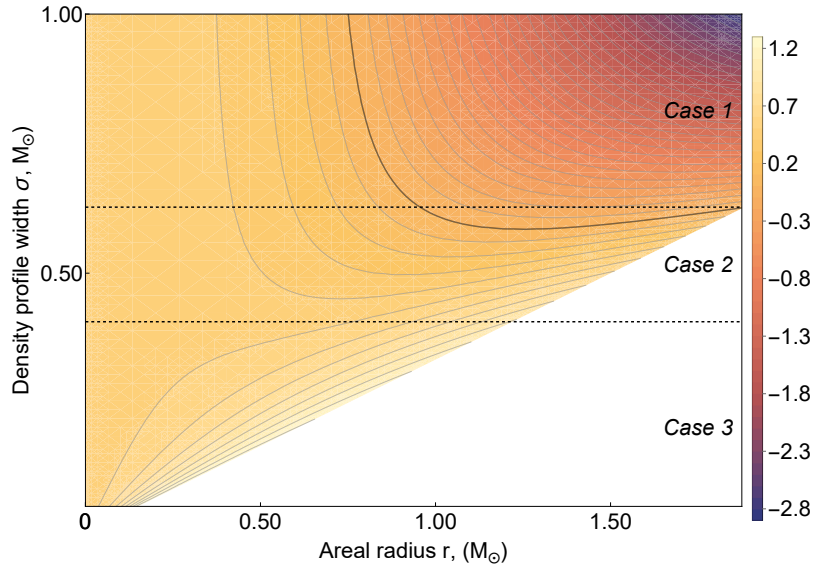


Figure 5.21: Contour plot of $t_{\text{AH}}(r; r_b)$ for $\rho_c = 0.25 M_{\odot}^{-2}$. The black contour line is $t_{\text{AH}} = 0$. The black dashed lines demarcate the boundaries of r_b that separate each case in Fig. 5.20. The region $r > r_b$ is excised.

As we have seen, the qualitative analysis of linear and Gaussian inhomogeneous matter distributions yields nearly identical results. While the Gaussian profile introduces more complicated expressions and smoother data—owing to its physically motivated initial energy density—the overall collapse dynamics and the structure of apparent horizon formation remains unchanged. Both profiles exhibit the same qualitative insensitivity to parameter variations in the matter collapse and reveal the same threefold classification of collapse behavior across parameter space. In the absence of a counterexample or compelling theoretical argument to the contrary, we conjecture that any **monotonically decreasing** initial energy density profile will give rise to the same qualitative features observed in these two cases.

5.3 Asymptotics of $M(t, r)/r$ near the matter center

In order to understand and compare the nature of the singularity forming at the center of each collapsing model, it is essential to examine the asymptotic behavior of the metric function near $r = 0$. In particular, the ratio $M(t, r)/r$, which directly enters the spacetime metric, serves as a useful diagnostic for probing the central behavior. As mentioned in passing, $M(t, r)$ becomes multivalued beyond the critical time and is therefore excised from our analysis. To proceed, we instead analyze the asymptotic limit of $M(t, r)/r$ as $r \rightarrow 0$.

Our method involves curve-fitting near the origin. Specifically, we fit the quantity according to the relation

$$\lim_{r \rightarrow 0} \frac{M(t, r)}{r} \sim Ar^k, \quad (5.46)$$

where A and k are time-dependent constants determined on each time slice. When $k > 0$, the mass function vanishes faster than r , and thus $M(t, r)/r \rightarrow 0$ at the center, indicating regular behavior. In contrast, we identify a genuine singularity when $k < 0$, since $M(t, r)/r$ diverges as $r \rightarrow 0$, implying a blow-up of our metric functions at the center. This loss of regularity ruins the underlying manifold structure of our spacetime, and hence is what we define as geometric singularities. We compile the best-fit curves in Fig. 5.22.

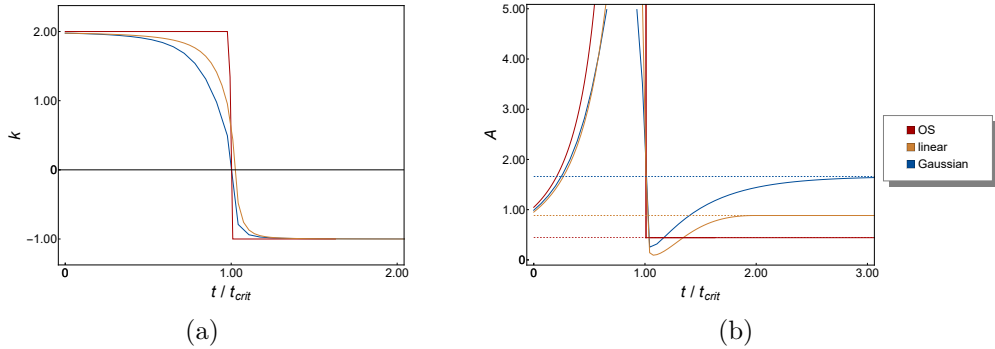


Figure 5.22: Comparison of best-fit curves for the asymptotic behavior of $M(t, r)/r$ as $r \rightarrow 0$, based on (5.46), for each matter configuration considered. All fits have R^2 values exceeding 0.97. The dashed colored lines indicate the respective total mass of each case.

All models agree on $k \approx 2$ at the initial time slice, indicating that $M_0 \sim r^3$, as expected. For the Oppenheimer–Snyder model, k remains at 2 throughout the

entire collapse process. Hence, the mass function behaves as $M(t, r) \sim Ar^3$ prior to the critical time. As shown in Fig. 5.22, the coefficient A steadily increases as the matter becomes increasingly localized. At the critical time, k abruptly shifts to -1 , signaling the formation of a genuine physical singularity. Specifically, $M(t, r)/r \sim A/r$, indicating that the mass at $r = 0$ becomes a finite constant, which coincides with the total mass of the star. This is consistent with the fact that all dust shells in the OS spacetime simultaneously collapse to $r = 0$ at the critical time.

For both inhomogeneous models, we observe the same overall behavior: the value of k becomes negative near or just past the critical time, again indicating the formation of a physical singularity. What differs in these cases is that the transition in k is more gradual, precisely reflecting the more staggered collapse of the dust shells in each case. We find that the best-fit curves of k for both the linear and Gaussian models asymptotically approach -1 , indicating that the mass function again tends toward a finite constant, which asymptotically matches the total mass of the respective star. As expected, since the linear model collapses more rapidly than the Gaussian one, both k and A approach their asymptotic values more quickly in the linear case.

As our next logical extension, we wish to immerse our local collapse dynamics onto an exterior FLRW universe. By doing so, we aim to investigate how the global expansion of the universe can affect, or perhaps even alter, the local collapse of an inhomogeneous matter distribution. This setup promises a more realistic cosmological context compared to the idealized vacuum exterior. As we will see in the next chapter, however, this matching introduces significant challenges and limitations within the formalism we have adopted.

Chapter 6

FLRW background

As a final model, we now wish to immerse our collapsing inhomogeneous star in an expanding Friedmann universe. This way, we will be able to probe how the global expanding universe can influence the local dynamics of the collapsing spacetime. Moreover, we wish to be able to recover the scales in which the global expansion decouples from the local collapse. To much dismay, however, we find that this spacetime matching is *not* allowed in this formalism.

To continuously match an LTB interior with an FLRW exterior at the boundary $r = r_b$, we are required to restrict our energy density to be only a function of time for $r > r_b$ i.e.

$$\rho(t, r > r_b) = \rho(t) \quad (6.1)$$

Therefore, in the region $r > r_b$ our field equations reduce to

$$\begin{aligned} \frac{d}{dt} \left(\frac{k}{a^2} \right) &= \frac{2k}{a^2} \sqrt{\frac{8\pi\rho}{3} + \frac{k}{a^2}} \\ \frac{\dot{a}}{a} &= -\sqrt{\frac{8\pi\rho}{3} + \frac{k}{a^2}} \end{aligned} \quad (6.2)$$

which is only half of the Friedmann equation. Specifically, this is the root that produces scale factor solutions that are strictly monotonically decreasing functions of time i.e. collapsing geometries. Hence, our formalism is limited to matching a collapsing LTB interior to a collapsing FLRW exterior. Note that this issue has propagated because of our explicit choice of shift vector during the derivation of the 3+1 solution at (4.8). In essence, this is a direct consequence of instead choosing the negative root of (3.8a) when matching the shift vectors at (4.19). Hence, the same issue will occur if we chose the positive root and that we will be limited to matching an expanding LTB to an expanding FLRW. In either case, we *cannot* construct a

locally collapsing spacetime embedded to a globally expanding spacetime with our choice of coordinates.

Regardless, we now attempt to solve the remaining PDE for completeness' sake, even if it is not entirely physical. Rewriting (4.29) here for convenience,

$$\frac{\partial M}{\partial t} - \sqrt{\frac{2M}{r}} \frac{\partial M}{\partial r} = \sqrt{2M} \, 4\pi P(t) r^{3/2}. \quad (6.3)$$

We again solve this using the method of characteristics, where this time our system of ODEs are:

$$\frac{\partial t}{\partial \tau} = 1 \quad t(0, x) = 0 \quad (6.4a)$$

$$\frac{\partial r}{\partial \tau} = -\sqrt{\frac{2\mu}{r}} \quad r(0, x) = x \quad (6.4b)$$

$$\frac{\partial \mu}{\partial \tau} = \sqrt{2\mu} \, 4\pi P(\tau) r^{3/2} \quad \mu(0, x) = M_0(x) \quad (6.4c)$$

Just as before $t(\tau, x) = \tau$, but now we no longer have a closed form solution for M . Moreover, (6.4c) dictates that the mass is no longer conserved along the characteristics. What we have now is a simultaneous system for r and M that we shall evolve across a discrete grid of initial points r_0 to determine the characteristics/geodesics¹ of our fluid elements and the evolving mass that each of them carries. Now, we have a free function $P(t)$ which we must determine to carry out our simultaneous system.

Recall that we choose our perfect fluid such that

$$\rho(t, r) = \rho_{\text{int}}(t, r) + \rho_b(t) \quad P_{\text{int}}(t) = 0 \quad \text{and} \quad P_b(t) = P(\rho_b) \quad (6.5)$$

and that in the background region, the coordinates describe an FLRW geometry. By choosing our equation of state to be a linear one describing a cosmic fluid $P(t) = \omega \rho_b(t)$, the energy density outside will evolve such that

$$\rho_b(t) = \rho_b \left(1 - t \sqrt{6\pi(1 + \omega)^2 \rho_b} \right)^{-2} \quad (6.6)$$

which we verified in the Oppenheimer-Snyder discussion. Hence, our pressure function would be

$$P(t) = \omega \rho_b \left(1 - t \sqrt{6\pi(1 + \omega)^2 \rho_b} \right)^{-2} \quad (6.7)$$

Substituting this to (6.4), our final simultaneous system is

$$\frac{\partial r}{\partial t} = -\sqrt{\frac{2M}{r}} \quad r(0) = r_0 \quad (6.8a)$$

¹From (4.4), we know that these are still geodesics.

$$\frac{\partial M}{\partial t} = \sqrt{2M} \, 4\pi\omega\rho_b \left(1 - t\sqrt{6\pi(1+\omega)^2\rho_b}\right)^{-2} r^{3/2} \quad M(0, r_0) = M_0(r_0) \quad (6.8b)$$

where we simply obtain the characteristics $r(t)$ originating from each point r_0 and the evolving mass along each characteristic $M_0(r_0)$. Note that we do not explicitly obtain the metric function $M(t, r)$. To do so, we must track the values of $M_0(r_0)$ as it evolves along each characteristic and obtain an interpolated function for M . However, we are no longer interested in this result and will forego this analysis. Because of this, we will also not be able to compute for the apparent horizons of our spacetime. Moving forward, we will only analyze the timelike geodesics to track the collapse of our spacetime. Furthermore, we posit that since an inhomogeneous distribution of dust produces a genuine black hole when it is immersed in a vacuum background, then a collapsing background must also necessarily cause it to produce a black hole.

As we have seen from the previous section, there is not much qualitative difference from considering an idealized linear inhomogeneous distribution from a more realistic Gaussian one. Since we are already considering an unrealistic scenario, we might as well consider a linearly decreasing initial energy density function with a background density and recover only the qualitative results from our analysis. We parametrize our energy density as

$$\rho_0(r) = \begin{cases} \rho_c \left(1 - \frac{r}{r_b}\right) + \rho_b & \text{if } r \leq r_b, \text{ and} \\ \rho_b & \text{if } r > r_b \end{cases} \quad (6.9)$$

Since we want the interior matter to be an overdensity, we impose $\rho_c > \rho_b$. We know that the evolution of the fluid elements do not depend with the parameters, and such we fix our parameters to $\rho_c = 0.25M_\odot^{-2}$ and $r_b = 0.25M_\odot$. Our initial mass function then is:

$$M_0(r) = 4\pi \times \begin{cases} \left(\frac{r^3}{3} - \frac{r^4}{4r_b}\right) \rho_c + \frac{\rho_b r^3}{3} & \text{if } r \leq r_b, \text{ and} \\ \frac{\rho_c r_b^3}{12} + \frac{\rho_b r^3}{3} & \text{if } r > r_b \end{cases} \quad (6.10)$$

and consider $M_s := M_0(r_b) = 4\pi(\rho_c/12 + \rho_b/3)r_b^3$ as the unperturbed conserved mass of the inhomogeneous matter configuration. We now consider two different exterior fluids for our analysis; a matter-dominated ($\omega = 0$) and a radiation-dominated ($\omega = 1/3$) one.

6.1 Matter-dominated collapse

As a first example, we impose $\omega = 0$ on the background homogeneous universe. In essence, we are studying a globally collapsing spacetime where we have explicitly introduced an overdensity. In this case, the system of ODEs (6.8) is the same system as (5.2) and we are able to provide a full analysis as we have done in the previous chapter. We skip the entire procedure and focus on the effects of the background FLRW on the collapse. In the vacuum background case, the innermost shell collapses into a singularity at time $t = t_{\text{crit}}$ where

$$t_{\text{crit}} = \frac{1}{\sqrt{6\pi\rho_0(r=0)}} = \frac{1}{\sqrt{6\pi(\rho_c + \rho_b)}} \quad (6.11)$$

and the entirety of the star completes its collapse at $t = 2t_{\text{crit}}$. Moreover, we expect the exterior collapsing FLRW to also have a finite collapse time $t_{\text{coll}} = 1/\sqrt{6\pi\rho_b}$, where even the dust shells from infinity will inevitably collapse into the singularity. We observe the collapse of the inhomogeneous dust for various ρ_b at Fig. 6.1.

It is evident that for small ρ_b the collapse scenario finalizes at around $t = 2t_{\text{crit}}$, but as the background density increases, the entire collapse process ends at earlier times. For background densities $\rho_b \geq \rho_c/3$, the global spacetime collapses too quickly that it collapses at $t \leq 2t_{\text{crit}}$, faster than the unperturbed inhomogeneous star would have been.

The other solution to our simultaneous system is the mass along each geodesic, which we illustrate at Fig. 6.2. Note that this is self-similar for all inputs of ρ_b , which is evident from (6.8b). It is apparent that the mass is in fact conserved.

6.2 Radiation-dominated collapse

For a lack of options to implement in our formalism, we choose $\omega = 1/3$ and classify this as a “radiation-dominated” collapse similar to the radiation-dominated epoch of the standard cosmology. We can think of this scenario happening in the late stages of a theoretical cosmological bounce, where an overdensity of dust is collapsing due to the influence of its self-gravity and by the background contraction of the radiation-dominated universe. The radiation component does not cluster (due to its pressure support) and remains homogeneous, while the dust, being non-relativistic, clumps and collapses more rapidly.

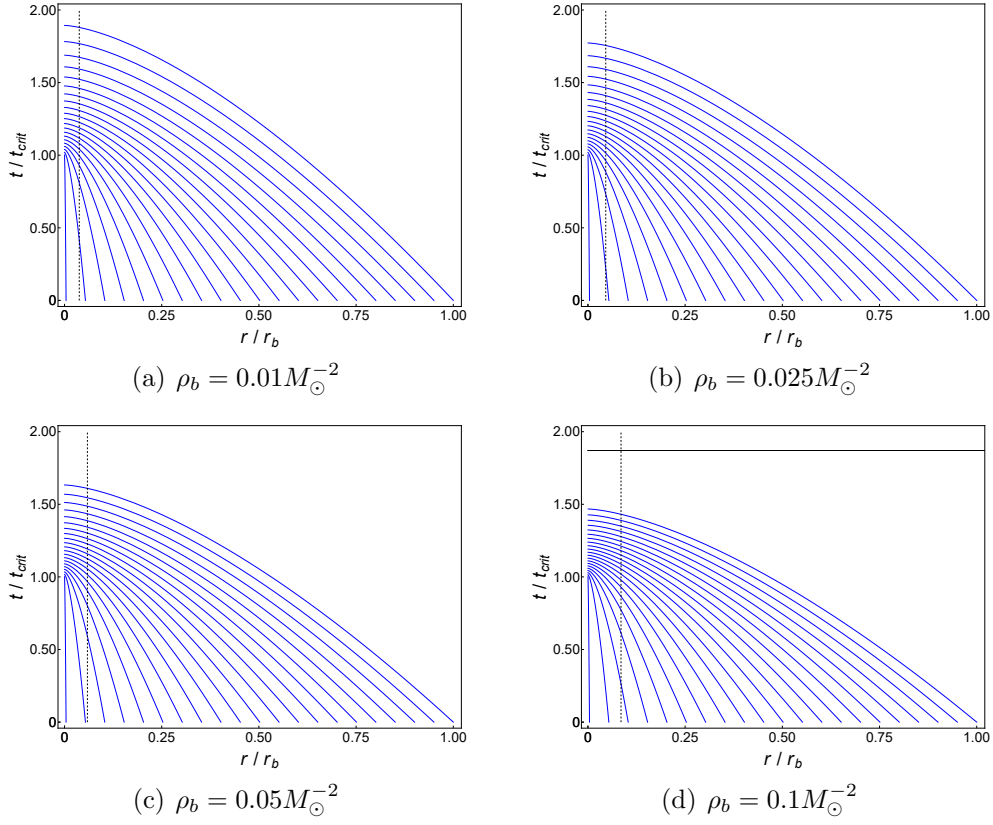


Figure 6.1: Timelike geodesics of the inhomogeneous star immersed in a matter-dominated collapsing FLRW universe. The dashed line is $r_s = 2M_s$. The solid line on (d) is t_{coll} .

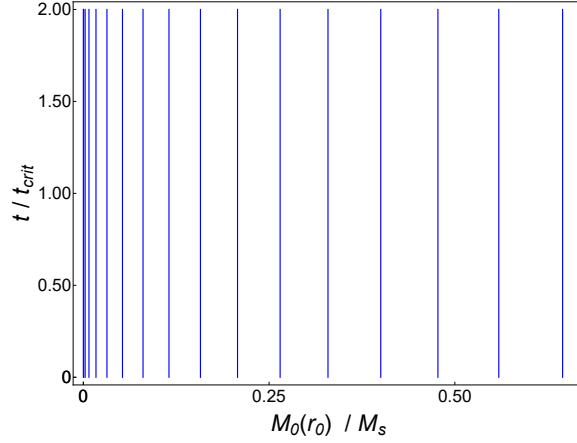


Figure 6.2: Mass evolution along the geodesics of the matter-dominated LTB-FLRW matched spacetime

The timelike geodesics are shown in Figure 6.3, where the critical time is kept fixed at (6.11). As in the matter-dominated background, the collapse proceeds more rapidly than in the unperturbed case. The presence of the exterior radiation component

accelerates the collapse of the interior dust, as evidenced by the slight difference in the collapse time of the star's outermost shell between the two cases.

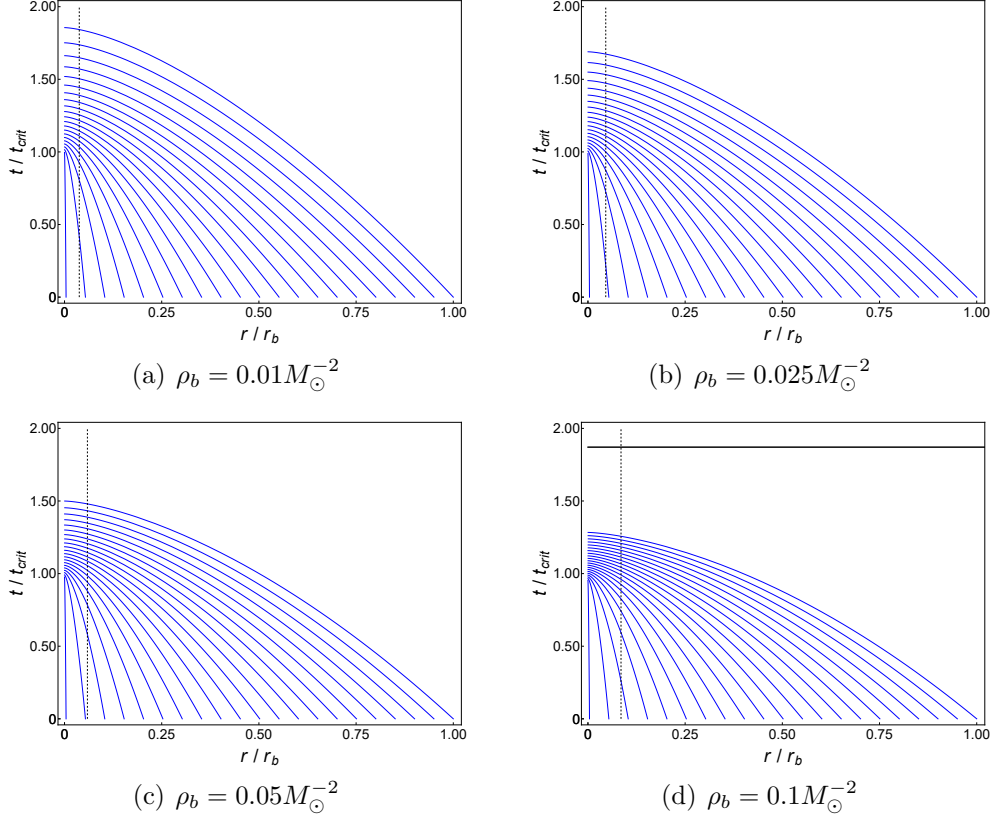


Figure 6.3: Timelike geodesics of the inhomogeneous star immersed in a radiation-dominated collapsing FLRW universe. The dashed line is $r_s = 2M_s$. The solid line on (d) is t_{coll} .

The mass along each geodesic is shown in Fig. 6.4. We can clearly see that the background universe imparts mass onto the freely-falling fluid shells, which is proportional to the density of the background.

As a final comment, we note that our analysis here was limited by our choice of numerical method for solving the field equation. Because we employed the method of characteristics, we were restricted to probing only the timelike geodesics (characteristic curves) of our spacetime. We later discovered that the full partial differential equation can be solved using the method of lines, allowing us to compute a complete solution for $M(t, r)$ on a finite computational grid. With this approach, we would then be able to extend our analysis and study the model in the same manner as before. Moreover, we can readily use the current formalism for the matter-dominated background case, as imposing $P(t) = 0$ will simply lead us to the same field equa-

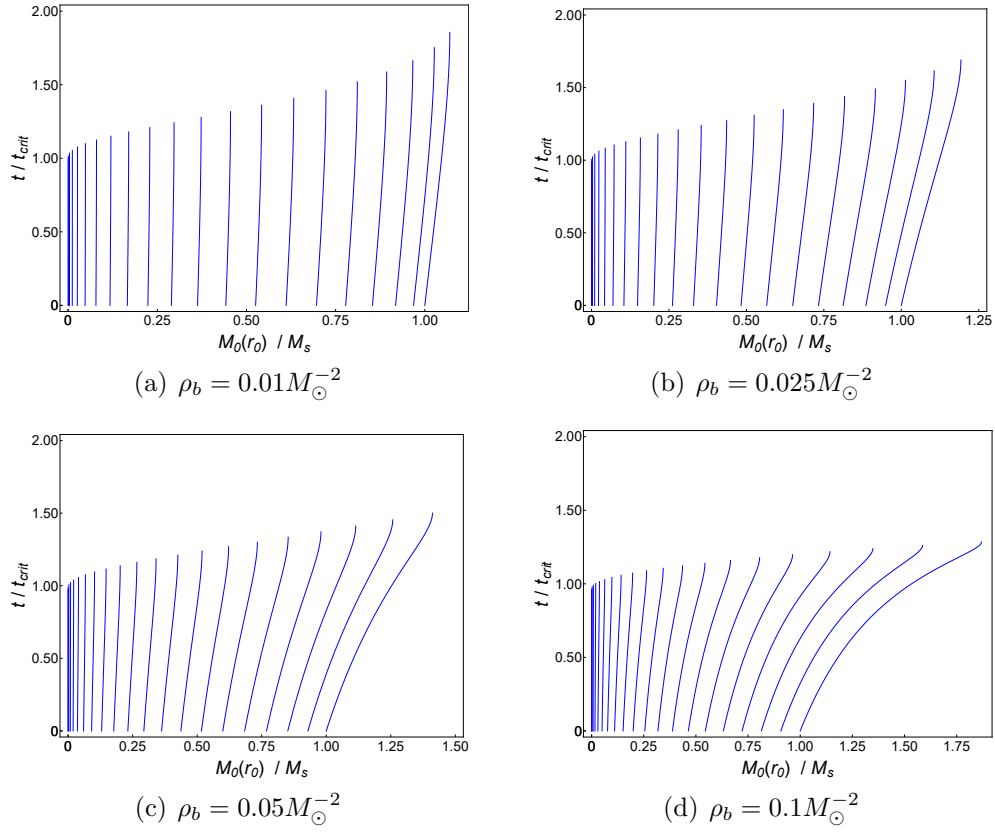


Figure 6.4: Mass evolution along the geodesics of the radiation-dominated LTB-FLRW matched spacetime

tion that we have treated fully in Chapter 5. We have preliminary results for this treatment, but we shall leave this as well as the analysis of the full solution using the method of lines to future work.

Chapter 7

Summary and conclusions

In summary, we have analytically and numerically studied the collapse of various cases of inhomogeneous dust using Painlevé–Gullstrand coordinates. By constructing the spacetime using a single coordinate patch, we have avoided the technical and conceptual complications associated with junction conditions. Moreover, these coordinates are horizon-penetrating, allowing the geometry of collapse to be tracked smoothly beyond the formation of apparent horizons. This is an essential feature for probing the internal structure of black holes and the final stages of stellar evolution. This makes the framework particularly well-suited for numerical implementations, where regularity and stability are critical across evolving domains.

A central finding of this work is that incorporating inhomogeneity in the matter source delays the collapse and introduces interesting behavior in apparent horizon formation. We further conjecture that this qualitative result holds regardless of the specific functional form of the inhomogeneity in the initial energy density. In choosing the matter sources to evolve, we parameterized the initial energy density using the central density ρ_c and the star boundary r_b . We find that for each specific parametrization, the collapse of matter from its initial configuration into a singularity does not qualitatively depend on the input parameters; all scenarios are equivalent up to some scaling factors. However, the formation of apparent horizons is sensitive to the choice of parameters.

For the exterior vacuum cases, we have obtained a full characterization of all possible collapse outcomes by probing the parameter space of (ρ_c, r_b) . Additionally, we analytically derive an expression for the time it takes the i^{th} dust shell to reach its Schwarzschild radius, complementing the parameter space analysis and providing further insight into the collapse process.

We have also found that immersing local collapsing matter into a collapsing external universe accelerates the collapse. However, due to constraints in the numerical method employed, our analysis of the exterior collapsing FLRW case is limited to tracking timelike geodesics (i.e., characteristic curves) to observe the behavior of the collapsing matter. As previously noted, the method of lines can be readily applied to access the full solution for the mass function and extend to our standard analysis. We leave these results to future work.

7.1 Recommendations

A natural extension of this work is to embed the local collapsing matter distribution within a globally expanding universe. This would enable us to investigate the interplay between local gravitational collapse and cosmic expansion, and to explore how the global dynamics of the universe influence local structure formation. In particular, we aim to identify a characteristic length scale beyond which cosmic expansion dominates and within which local contraction prevails—an idea motivated by the observed behavior of local galaxy clusters.

However, our analysis suggests that achieving this separation exactly within our framework may not be feasible. The evolution of our spacetime is governed by a binary choice in the radial component of the shift vector, effectively constraining the spacetime to be either entirely collapsing or entirely expanding. This limitation prevents us from modeling scenarios where collapse and expansion coexist within different regions of the same spacetime. As a result, a full nonlinear treatment of the local-global dynamics appears unattainable in this setting, and we are compelled to adopt perturbative methods, as is commonly done in the literature.

Another natural extension of this work is to include radial pressure gradients in order to investigate structure formation processes leading to stars and galaxies. This requires solving the full Einstein field equations given in (4.9), specifically allowing for nonzero pressure gradients such that $E \neq 0$. In their original study, Lasky and Lun [10] demonstrated that the full system admits the well-known Tolman-Oppenheimer-Volkoff (TOV) solution in the static limit, which describes matter in hydrostatic equilibrium. Therefore, a complete solution for the metric functions $M(t, r)$ and $E(t, r)$ should enable us to distinguish between configurations that lead to gravitational collapse and black hole formation, and those that stabilize into non-singular

structures.

Since the system was derived using the ADM formalism, we believe that an appropriate gauge choice exists in which the evolution equations are well-posed, yielding stable and convergent solutions to the hyperbolic system of partial differential equations. Further work is needed to reformulate or simplify the full system to uncover such a gauge and ensure numerical stability for simulations involving pressure-supported matter configurations.

Appendix A

Conventions and implementation

Throughout this text, we adopt a +2 Lorentzian signature for the metric tensor. We shall also use the abstract index notation, which differentiates the tensor g_{ab} (labeled by Latin letters) from its components $g_{\alpha\beta}$ (labeled by Greek letters). These indices run from 0 to 3, while the middle Latin letters $i, j, k, \dots$ will run from 1 to 3, usually reserved for spatial indices. We adopt the Einstein summation convention of summing repeated indices. Lastly, we shall use geometrized units and set the gravitational constant and speed of light to unity, i.e. $G = c = 1$. This choice leaves all physical variables to be in units of length, and we shall use the solar mass $M_{\odot} \approx 1.5 \text{ km}$ as the characteristic length scale for our simulations later on¹.

Since our analysis followed a systematic pipeline, we include the code used to generate the plots for the linear inhomogeneous case here², along with its generalization for producing the parameter space analysis plot, which aggregates the results of 400 simulations. We invite the reader to explore and experiment with their own initial matter configurations using this framework.

¹This is also available at the start of Chapter 2

²For physical readers, <http://tiny.cc/k3z1001>.

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